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THE ROYAL SOCIETY

Generalized Markowitz mean-variance principles for multi-period portfolio-selection problems

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The multi-period portfolio-selection problem is formulated as a Markowitz mean-variance optimization problem. It is shown that the single-period Markowitz quadratic programming algorithm can be used to solve the multi-period asset-allocation problem with suitable modifications in the covariance and linear constraint matrices. It is assumed that the number of shares invested in risky assets is deterministic and the amount of money invested in the risk-free asset is random at future trading dates when short sales on assets are allowed. The general covariance matrix in the multi-period setup contains intertemporal correlations between assets, in addition to correlations between assets at all trading dates. Analytical solutions for the optimal trading strategy, which is linear in the risk-aversion parameter, are obtained. The efficient frontier is a straight line in the expected return/standard deviation of the portfolio space. When the dynamics of the risky assets follow geometric Brownian motion, it is shown that the time-zero allocations to the risky assets coincide with those obtained by Merton in the continuous-time framework. When short sales are not allowed on assets, the values of the portfolio at future trading dates may not be conserved. In the modified Markowitz mean-variance formulation, the value of the portfolio at future trading dates is conserved in the expected sense, and the optimal trading strategy is selected so that the deviation is minimized in the least-square sense. The efficient frontier is a parabola in the expected return/total variance space when short-sales are allowed and when the portfolio consists of risky assets only. When short sales are not allowed, it is shown that finding the optimal trading strategy is equivalent to solving the single-period Markowitz quadratic programming problem, with suitable modifications in the covariance and linear constraint matrices. By solving the Karush–Kuhn–Tucker conditions, analytical solutions are obtained for a two-period two-assets case, and the Markowitz mean-variance principle is illustrated by solving a test problem numerically using the revised simplex method.

Keywords: multi-period portfolio-selection problem; Markowitz mean-variance principle; Karush–Kuhn–Tucker conditions; quadratic programming

1. Introduction

In his seminal work, Markowitz (1952, 1987, 1994) illustrated how the mean-variance principle is used to generate the admissible subset of investments for risk-averse indi-

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viduals by eliminating any investment that has a lower mean and higher variance than a member of the given set of investment alternatives. The mean-variance principle makes use of the first two moments that characterize the random behaviour of the risky assets, and it seeks to determine portfolios of securities that are efficient with respect to expected return and variance, which include both up and down deviations from the expected return of the portfolio. When semivariance, which counts only downward deviations and not up and down deviations as does variance, is estimated from historical data, Markowitz *et al.* (1993) have shown that the mean-semivariance problem can be formulated as a mean-variance optimization problem. Despite all the objections, the mean-variance approach has maintained its popularity because it provides a tractable theoretical basis for portfolio selection and evaluation, and it is typically much more economical to trace out a mean-variance efficient set than to maximize the expected utility of terminal wealth. For example, depending on the utility function and the form of the joint distribution for the assets assumed, maximization of the expected utility of terminal wealth for a few securities can be as expensive as tracing out a mean-variance efficient set for thousands of securities. The simplicity lies in obtaining closed-form solutions when there are no inequality constraints on the variables. The crux of the mean-variance principle lies in the fact that, when linear inequality constraints are present, the optimal solution is a piecewise linear function of the risk-aversion parameter, known as the critical lines, which are the solutions to certain affine portfolio-selection problems related to the original problem. Karush–Kuhn–Tucker (KKT) conditions are invoked to find the bounds for the risk-aversion parameter. Corresponding to each interval for the risk-aversion parameter, the affine portfolio-selection problem depends upon the inequality constraints that are active. For the general mean-variance theory and its numerical implementation, we refer to Markowitz (1987) and Best (1996, 1999). Best & Grauer (1991*a*, *b*) have developed analytical results for the sensitivity of the optimal portfolio to changes in the mean returns of the assets and the source vector in the constraints.

The mean-variance principle addresses the investor's asset-allocation problem for an investment horizon of one period. Ziemba & Vickson (1975) have collected numerous classic articles on both static and dynamic models of portfolio selection. Recently, Ziemba & Mulvey (1998) have edited a collection of articles on asset and liability modelling. There are three general frameworks for finding optimal decisions over the planning horizon. They are (a) stochastic optimal control theory, (b) capital growth theory and (c) the stochastic programming approach.

Stochastic control theory is a general-purpose framework; it applies to problems in which the state-space can be kept manageable, that is, with at most three or four. Using Ito's lemma and Bellman's dynamic programming, Merton (1969, 1996) has shown how to construct and analyse optimal continuous-time allocation problems under uncertainty. Merton's model has been generalized in several directions (Breeden 1979; Cox *et al.* 1985; Brennan *et al.* 1997). Ingersoll (1987) has presented a detailed overview of these models. The introduction of proportional transaction costs to Merton's model has been studied by Magill & Constantinides (1976), Davis & Norman (1990), Shreve & Soner (1994) and Akian *et al.* (1996), among others. The problem has been reduced to solving the Hamilton–Jacobi–Bellman equation, a parabolic partial differential equation, for the value function, and its use is limited by the number of assets in the portfolio.

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Capital growth theory (Hakansson 1974; Hakansson & Ziemba 1995) shows how to maximize the long-run growth of assets without borrowing or liabilities. Under certain assumptions, this is done by maximizing the expected log of wealth. The long-term expected capital growth is maximized by selecting a strategy that maximizes the expected logarithm of a single return, and thus the capital growth strategy is myopic. Capital growth models lead to the high growth of assets, but with considerable risk. When controlled through fractional Kelly strategies or drawn from constraints, the policies provide superior growth security tradeoffs. The policies tend to concentrate on a small number of superior assets and hence can be poorly diversified. The capital growth model is well suited for wealthy individuals who are willing to accept considerable risks and thus caters to the needs of a small fraction of society. Hakansson & Ziemba (1995) have reviewed the capital growth literature.

A stochastic program determines the optimal investment at each period as a function of the outlook for the assets in comparison with the investor's circumstances using a large multi-stage stochastic program. The possible sequence of outcomes can be represented as an event tree, in which branchings occur at each period; each branch represents one possible outcome over one period, and each path in the tree represents one possible sequence of outcomes, i.e. one scenario. The number of unknown allocation variables is of the order Nb^M, where N denotes the number of assets in the portfolio, b the branching factor (number of possible outcomes for each asset's return in each period) and M the number of periods under consideration. Significant applications of stochastic programming include Carino *et al.* (1994), who describe a case study of a Japanese insurance company with 256 scenarios, with eight branches from the initial node and two or three thereafter, Mulvey (1994), who analyses an asset-liability problem, Mulvey & Vladimirou (1992), who analyse a variety of financial planning problems with up to eight periods and 100 scenarios, and Zenios (1993), who develops a model for the management of a portfolio of mortgage-backed securities. Consigli & Dempster (1998) have presented the computer-aided asset/liability management (CALM) stochastic programming model for dynamic asset-liability management. The number of periods and the need for the optimal trading strategy for different risk-aversion parameters limits this approach.

The aim of this paper is to formulate the multi-period portfolio-selection problem as a Markowitz mean-variance optimization problem. The main objectives are to obtain analytical solutions for the optimal trading strategy when there are no restrictions on short sales and to show that the existing quadratic programming algorithms can be used to solve the multi-period asset-allocation problems with suitable modifications in the covariance and constraint matrices when there are short-sale restrictions on assets. In § 2, the dynamic asset-allocation problem is formulated when the portfolio consists of a risk-free asset. In § 3, using the Markowitz mean-variance principle, analytical solutions are obtained when there are no restrictions on short sale of assets. Some particular cases considered include those in which the dynamics of the risky assets follow correlated geometric Brownian motion and in which there is no intertemporal dependence on asset prices. In § 4, the asset-allocation problem is formulated in terms of the generalized Markowitz mean-variance principle, when the portfolio need not contain a risk-free asset. The values of the portfolios at future trading dates are not conserved exactly. They are conserved in the expected sense, and the deviation is minimized in the least-square sense. Analytical solutions and efficient frontiers are obtained for the cases when the portfolio contains only risky

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assets and when the portfolio contains a risk-free asset. Section 5 deals with the portfolio-selection problem when there are non-negative constraints on asset allocations. Using the KKT conditions, analytical solutions are obtained for a test problem when the portfolio has one risky asset and one risk-free asset, and when the number of trading periods is two. All possible price movements of the risky asset with respect to the risk-free asset are considered. Using the quadratic programming algorithm, the mean-variance principle is illustrated by numerically solving a test problem when the portfolio consists of seven assets, and when the number of trading periods is ten. In § 6, the implications of the open-loop policy for the trading strategy on the results are discussed. Section 7 concludes the paper.

2. Formulation

We introduce the following notation:

$M+2$	total number of discrete time nodes;
$N+1$	total number of assets;
t_j	trading date, $j = 0, 1, \ldots, M$;
X	available wealth at time $t = 0$;
$S_{i,j}$	price of the ith asset at $t = t_j$;
$U_{i,j}$	number of shares of the ith asset held at $t = t_j$ after rebalancing.

The assets are bought at time $t = t_0 = 0$ with the available wealth X. This leads to the budget constraint given by

$$\sum_{i=0}^{N} S_{i,0} U_{i,0} = X. \tag{2.1}$$

We assume that rebalancing takes place instantaneously at discrete times t_j, for $j = 1, 2, \ldots, M$. The value of the portfolio $W(t_j)$ at $t = t_j$ before the transactions is given by

$$W(t_j) = \sum_{i=0}^{N} S_{i,j} U_{i,j-1}, \tag{2.2}$$

and the value of the portfolio $W'(t_j)$ at $t = t_j$ after the transactions is given by

$$W'(t_j) = \sum_{i=0}^{N} S_{i,j} U_{i,j}. \tag{2.3}$$

The conservation of wealth at any trading date $t = t_j$ leads to

$$\sum_{i=0}^{N} S_{i,j}(U_{i,j} - U_{i,j-1}) = 0, \quad j = 1, 2, \ldots, M. \tag{2.4}$$

At the end of the investment period $t = t_{M+1}$, the value of the portfolio $W(t_{M+1})$ is given by

$$W(t_{M+1}) = \sum_{i=0}^{N} S_{i,M+1} U_{i,M}. \tag{2.5}$$

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The time-zero prices of the assets $S_{i,0}$, $i = 0, 1, \ldots, N$, are deterministic, and future prices of the assets $S_{i,j}$, $i = 0, 1, \ldots, N$, $j = 1, 2, \ldots, M+1$, are random from the time-zero perspective. The number of shares bought at time zero, denoted by $U_{i,0}$, $i = 0, 1, \ldots, N$, is deterministic, whereas the number of shares to be bought at future trading dates, denoted by $U_{i,j}$, $i = 0, 1, \ldots, N$, $j = 1, 2, \ldots, M$, is stochastic from the time-zero perspective. Our goal is to find the trading strategy,

$$\{U_{i,j},\ i = 0, 1, \ldots, N,\ j = 0, 1, \ldots, M\},$$

that maximizes an expected objective function based on the terminal wealth given by (2.5) subject to the budget constraint (2.1) and the intermediate wealth conservation equations given by (2.4).

(*a*) *Assumptions*

In this paper we assume the following.

(a) The number of shares invested in each risky asset is deterministic at all times.

(b) The amount of money invested in the risk-free asset is random at future trading dates t_j, $j = 1, 2, \ldots, M$, when short sales are allowed.

(c) The amount of money invested in the risk-free asset is deterministic at all times when short sales are not allowed.

Due to assumption (a), the amount of money invested in each risky asset at future trading dates is random. The value of the portfolio at all times will be conserved exactly due to assumptions (a) and (b) when short sales are allowed and when the portfolio has a risk-free asset. When short sales are not allowed, the value of the portfolio at future trading dates may not be conserved exactly. In the next section, we study the asset-allocation problem when there is a risk-free asset and when there are no restrictions on short sales of assets.

(*b*) *One risk-free asset (stochastic control) and N risky assets (deterministic control), short sales allowed*

In this section, we study the multi-period asset-allocation problem when there is a risk-free asset identified by the index $i = 0$. Rising interest rates are normally associated with falling equity returns, and we assume that the short-term interest rate is not affected by asset prices (Spiro 1989). We assume that the price behaviour of the risk-free asset $\{S_{0,0} = 1, S_{0,1}, \ldots, S_{0,M+1}\}$ is known deterministically, and these values act as reference prices. The terminal wealth $W(t_{M+1})$, given by (2.5), can be written as

$$\frac{W(t_{M+1})}{S_{0,M+1}} = U_{0,M} + \sum_{i=1}^{N} \frac{S_{i,M+1}}{S_{0,M+1}} U_{i,M}, \tag{2.6}$$

where $W(t_{M+1})/S_{0,M+1}$ is the value of the portfolio discounted back to time zero. Similarly, wealth-conservation equations (2.4) can be written as

$$U_{0,j} = U_{0,j-1} - \sum_{i=1}^{N} \frac{S_{i,j}}{S_{0,j}} (U_{i,j} - U_{i,j-1}), \quad j = 1, 2, \ldots, M. \tag{2.7}$$

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Summing up over all equations of type (2.7) and using (2.1), we get

$$U_{0,M} = \frac{X}{S_{0,0}} - \sum_{i=1}^{N} \frac{S_{i,0}}{S_{0,0}} U_{i,0} - \sum_{j=1}^{M} \sum_{i=1}^{N} \frac{S_{i,j}}{S_{0,j}} (U_{i,j} - U_{i,j-1}). \tag{2.8}$$

We define the rate of net profit of the investment discounted back to time zero, R, by

$$R \equiv \frac{1}{X} \left\{ \frac{W(t_{M+1})}{S_{0,M+1}} - \frac{X}{S_{0,0}} \right\}. \tag{2.9}$$

Substituting (2.8) into (2.6) and using (2.9), we get

$$R = \frac{1}{X} \left\{ - \sum_{i=1}^{N} \frac{S_{i,0}}{S_{0,0}} U_{i,0} - \sum_{j=1}^{M} \sum_{i=1}^{N} \frac{S_{i,j}}{S_{0,j}} (U_{i,j} - U_{i,j-1}) + \sum_{i=1}^{N} \frac{S_{i,M+1}}{S_{0,M+1}} U_{i,M} \right\}. \tag{2.10}$$

Let $\alpha_{i,j}$ denote the number of shares of the ith risky asset transacted at time $t = t_j$, that is,

$$U_{i,j} - U_{i,j-1} = \alpha_{i,j}, \quad i = 1, 2, \ldots, N, \quad j = 1, 2, \ldots, M. \tag{2.11}$$

If the ith asset is bought (sold) at a trading date $t = t_j$, $\alpha_{i,j}$ is positive (negative). Summing up over all equations of type (2.11), we get

$$U_{i,M} = U_{i,0} + \sum_{j=1}^{M} \alpha_{i,j}, \quad i = 1, 2, \ldots, N. \tag{2.12}$$

Substituting (2.12) into (2.10), the rate of profit R can be written as

$$R = \frac{1}{X} \left[\sum_{i=1}^{N} \left\{ \frac{S_{i,M+1}}{S_{0,M+1}} - \frac{S_{i,0}}{S_{0,0}} \right\} U_{i,0} + \sum_{j=1}^{M} \sum_{i=1}^{N} \left\{ \frac{S_{i,M+1}}{S_{0,M+1}} - \frac{S_{i,j}}{S_{0,j}} \right\} \alpha_{i,j} \right]. \tag{2.13}$$

Defining $\alpha_{i,0} = U_{i,0}$, the number of shares of the ith asset bought at $t = 0$, equation (2.13) can be rewritten as

$$R = \frac{1}{X} \sum_{j=0}^{M} \sum_{i=1}^{N} p_{i,j} \alpha_{i,j}, \tag{2.14}$$

where

$$p_{i,j} = \frac{S_{i,M+1}}{S_{0,M+1}} - \frac{S_{i,j}}{S_{0,j}}, \quad i = 1, 2, \ldots, N, \quad j = 0, 1, \ldots, M, \tag{2.15}$$

is the difference between discounted asset prices at times $t = t_{M+1}$ and $t = t_j$. In the next section, we briefly illustrate how the portfolio-selection problem is solved using the expected utility maximization approach and the issues we encounter and, in the subsequent sections, we solve the asset-allocation problem using the Markowitz mean-variance principle.

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(*c*) *Expected utility maximization*

Using (2.9) and (2.14), the discounted terminal wealth can be written as

$$W_{M+1} \equiv \frac{W(t_{M+1})}{S_{0,M+1}} = \frac{X}{S_{0,0}} + \sum_{j=0}^{M}\sum_{i=1}^{N} p_{i,j}\alpha_{i,j}. \tag{2.16}$$

The trading strategy $\{\alpha_{i,j},\ j = 0,1,\ldots,M,\ i = 1,2,\ldots,N\}$ is obtained by maximizing the expected utility of the discounted terminal wealth, that is, by maximizing

$$F = E[J(W_{M+1})],$$

where E denotes the expectation operator. The utility function J is monotonically increasing and concave in its argument for a risk-aversion investor. The first-order optimality condition leads to

$$E\left\{\frac{\mathrm{d}J}{\mathrm{d}W_{M+1}}\left[\frac{S_{k,M+1}}{S_{0,M+1}} - \frac{S_{k,l}}{S_{0,l}}\right]\right\} = 0, \quad k = 1,2,\ldots,N, \quad l = 0,1,\ldots,M, \tag{2.17}$$

which, in turn, can be written as

$$\frac{S_{k,0}}{S_{0,0}} = E\left\{\frac{1}{E[\mathrm{d}J/\mathrm{d}W_{M+1}]}\frac{\mathrm{d}J}{\mathrm{d}W_{M+1}}\frac{S_{k,j}}{S_{0,j}}\right\}, \quad j = 1,2,\ldots,M+1, \quad k = 1,2,\ldots,N. \tag{2.18}$$

Equation (2.17) states that there exist states $\omega_{\mathrm{u}}^{k,l}$ and $\omega_{\mathrm{d}}^{k,l}$ so that the no-arbitrage condition

$$\frac{S_{k,M+1}}{S_{k,l}}(\omega_{\mathrm{d}}^{k,l}) < \frac{S_{0,M+1}}{S_{0,l}} < \frac{S_{k,M+1}}{S_{k,l}}(\omega_{\mathrm{u}}^{k,l})$$

holds good. Generally, the optimal trading strategy is obtained by solving the non-linear equations given by (2.17) or by maximizing the expected terminal wealth by the traditional Bellman's dynamic programming approach. The procedure has to be repeated for each risk-aversion parameter and this limits the number of assets and the number of trading dates. In what follows, we obtain closed-form solutions by setting the optimization problem in the Markowitz mean-variance framework when there are no inequality constraints on the trading strategy, and in the subsequent sections, we derive expressions for optimal trading strategy when linear inequality constraints are present.

3. Markowitz mean-variance principle

Let $\mu_{i,j}$ denote the expected price of the ith asset at time $t = t_j$, that is,

$$\mu_{i,j} = E\{S_{i,j}\}, \quad i = 1,2,\ldots,N, \quad j = 1,2,\ldots,M+1.$$

Taking the expectation of (2.14), we get

$$E(R) \equiv \mu_R = \frac{1}{X}\sum_{j=0}^{M}\sum_{i=1}^{N}\bar{p}_{i,j}\alpha_{i,j}, \tag{3.1}$$

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where $\bar{p}_{i,j} = E(p_{i,j})$. Let the N-dimensional vector $\alpha_j = (\alpha_{1,j}, \alpha_{2,j}, \ldots, \alpha_{N,j})^{\mathrm{T}}$, $j = 0, 1, \ldots, M$, denote the number of shares transacted at time $t = t_j$, and each N-dimensional vector

$$R_j = \left(\frac{\mu_{1,M+1}}{S_{0,M+1}} - \frac{\mu_{1,j}}{S_{0,j}}, \frac{\mu_{2,M+1}}{S_{0,M+1}} - \frac{\mu_{2,j}}{S_{0,j}}, \ldots, \frac{\mu_{N,M+1}}{S_{0,M+1}} - \frac{\mu_{N,j}}{S_{0,j}} \right)^{\mathrm{T}}, \quad j = 0, 1, \ldots, M+1,$$

denote the differences between discounted expected asset prices at $t = t_{M+1}$ and $t = t_j$. The vector $B = (\alpha_0^{\mathrm{T}}, \alpha_1^{\mathrm{T}}, \ldots, \alpha_M^{\mathrm{T}})^{\mathrm{T}}$ contains the unknown vectors of the number of shares transacted at times $t_0, t_1, \ldots, t_M$, respectively, and the vector $\mu = (R_0^{\mathrm{T}}, R_1^{\mathrm{T}}, \ldots, R_M^{\mathrm{T}})^{\mathrm{T}}$ denotes the vector of returns. With this notation, equation (3.1) can be written as

$$\mu_R = \frac{1}{X} \mu^{\mathrm{T}} B. \tag{3.2}$$

Similarly, the variance of R, denoted by σ_R^2, can be written as

$$\sigma_R^2 = \frac{1}{X^2} \sum_{j,n=0}^{M} \sum_{i,k=1}^{N} C_{j,n}(i,k) \alpha_{i,j} \alpha_{k,n}, \tag{3.3}$$

where $C_{j,n}(i,k)$ is the (i,k)th element of an $N \times N$ covariance matrix $C_{j,n}$ defined by

$$C_{j,n}(i,k) = E\{(p_{i,j} - \bar{p}_{i,j})(p_{k,n} - \bar{p}_{k,n})\}. \tag{3.4}$$

Defining

$$C = \begin{pmatrix} C_{0,0} & C_{0,1} & \cdots & C_{0,M} \\ C_{1,0} & C_{1,1} & \cdots & C_{1,M} \\ \vdots & \vdots & \ddots & \vdots \\ C_{M,0} & C_{M,1} & \cdots & C_{M,M} \end{pmatrix}, \tag{3.5}$$

equation (3.3) can be written as

$$\sigma_R^2 = \frac{1}{X^2} B^{\mathrm{T}} C B. \tag{3.6}$$

The matrix C, which contains the block matrices $C_{i,j}$ as elements, has $(M+1)N$ rows and columns. It can be shown that, for $j = 1, 2, \ldots, M$,

$$\left.\begin{aligned} C_{0,0} &= \frac{1}{S_{0,M+1}^2} \mathrm{Cov}_{M+1,M+1}, \\ C_{j,0} = C_{0,j}^{\mathrm{T}} &= \frac{1}{S_{0,M+1}^2} \mathrm{Cov}_{M+1,M+1} - \frac{1}{S_{0,j} S_{0,M+1}} \mathrm{Cov}_{j,M+1}, \end{aligned}\right\} \tag{3.7}$$

and for $j = 1, 2, \ldots, M$, $n = 1, 2, \ldots, M$,

$$\begin{aligned} C_{j,n} = \frac{1}{S_{0,M+1}^2} \mathrm{Cov}_{M+1,M+1} &+ \frac{1}{S_{0,j} S_{0,n}} \mathrm{Cov}_{j,n} \\ &- \frac{1}{S_{0,j} S_{0,M+1}} \mathrm{Cov}_{j,M+1} - \frac{1}{S_{0,M+1} S_{0,n}} \mathrm{Cov}_{M+1,n}, \end{aligned} \tag{3.8}$$

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where $\mathrm{Cov}_{j,n}(i,k)$ is the (i,k)th element of an $N \times N$ covariance matrix $\mathrm{Cov}_{j,n}$ defined by

$$\mathrm{Cov}_{j,n}(i,k) = E\{(S_{i,j} - \mu_{i,j})(S_{k,n} - \mu_{k,n})\}.$$

The matrices $C_{j,j}$, $j = 0, 1, \ldots, M$, are symmetric and $C_{n,j} = C_{j,n}^{\mathrm{T}}$, and hence C is symmetric positive-semi-definite.

In terms of the Markowitz mean-variance principle, our goal is to minimize

$$F = -\theta \frac{1}{X} \mu^{\mathrm{T}} B + \frac{1}{2} \frac{1}{X^2} B^{\mathrm{T}} C B, \tag{3.9}$$

where θ denotes the parameter that defines the risk aversion of the investor. In general, the parameter θ depends upon the initial wealth X of the investor, and hence is a function of X. It is to be noted that the parameter θ is different from the parameter θ', which appears on the optimization problem that minimizes

$$F' = -\theta' E\left\{ \frac{W(t_{M+1})}{S_{0,M+1}} \right\} + \tfrac{1}{2} \mathrm{Var}\left\{ \frac{W(t_{M+1})}{S_{0,M+1}} \right\},$$

though θ and θ' are related. In the above, Var denotes the variance operator. The first-order necessary condition for optimality to (3.9) leads to

$$CB = \theta X \mu. \tag{3.10}$$

If C is positive-definite, the optimal solution is given by

$$B = \theta X C^{-1} \mu. \tag{3.11}$$

Substituting (3.11) into (3.2) and (3.6), the optimal expected value of R and its variance are given by

$$\mu_R = \theta \mu^{\mathrm{T}} C^{-1} \mu, \qquad \sigma_R^2 = \theta^2 \mu^{\mathrm{T}} C^{-1} \mu.$$

These lead to the efficient frontier

$$\mu_R = \sigma_R \sqrt{\mu^{\mathrm{T}} C^{-1} \mu}, \tag{3.12}$$

which is a straight line in (μ_R, σ_R)-space. Using (2.9), the efficient frontier, in terms of the terminal wealth, can be written as

$$E\{W(t_{M+1})\} = \sigma_{W(t_{M+1})} \sqrt{\mu^{\mathrm{T}} C^{-1} \mu} + \frac{S_{0,M+1}}{S_{0,0}} X, \tag{3.13}$$

where $\sigma_{W(t_{M+1})}$ is the standard deviation of the terminal wealth. Writing

$$C^{-1} = (Q_0, Q_1, \ldots, Q_M)^{\mathrm{T}},$$

where each Q_i is a matrix of order $N \times (M+1)N$, equation (3.11) can be written as

$$\alpha_j = \theta X Q_j \mu, \quad j = 0, 1, \ldots, M. \tag{3.14}$$

The investment in the risk-free asset at time $t = 0$, using the budget constraint (2.1) and (3.14), is given by

$$U_{0,0} = \frac{1}{S_{0,0}} \{X - S_0^{\mathrm{T}} \alpha_0\} = \frac{X}{S_{0,0}} \{1 - \theta S_0^{\mathrm{T}} Q_0 \mu\}, \tag{3.15}$$

where $S_j = (S_{1,j}, S_{2,j}, \ldots, S_{N,j})^{\mathrm{T}}$ is the vector of prices of risky assets at time $t = t_j$. The parameter $\theta = \theta_M$ at which $U_{0,0} = 0$ is given by $\theta_M = 1/S_0^{\mathrm{T}} Q_0 \mu$, and hence it is imperative that $S_0^{\mathrm{T}} Q_0 \mu$ be positive. In this case, $U_{0,0}$ is a monotonically decreasing function of the parameter θ. Future investments in the risk-free asset, that is, $U_{0,j}$, $j = 1, 2, \ldots, M$, can be determined from (2.7), (3.14) and (3.15).

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(a) *The role of intertemporal correlations*

The general covariance matrix, defined by (3.5), contains intertemporal correlations between assets, in addition to correlations between assets at all trading dates. Time-varying expected returns and intertemporal correlations between assets, due to changing business conditions, generate some predictability in asset returns. This is one of the striking empirical findings in recent financial research (Summers 1986; Fama & French 1988; Poterba & Summers 1988; Lo & MacKinlay 1988; Lo 1991; Campbell *et al.* 1997).

The variances of stock price changes do not grow linearly with time as they would if a true random walk held. Instead, in short periods, they grow a bit faster than time, showing weak positive autocorrelation. That is, a positive return is more likely to be followed by another positive return than a reversal, and a negative return is more likely to be succeeded by another negative return than a positive return. For longer periods of time, the autocorrelation becomes significantly negative. In this mean reversion case, returns revert to an equilibrium value. If an asset is priced above its equilibrium value, it will be more inclined to decrease than to increase. Conversely, if an asset is priced below its equilibrium value, it will be more likely to increase than to depreciate further. According to Samuelson (1991, p. 198), 'the market is not necessary macroefficient; instead it undergoes self-fulfilling waves above and below fundamentalist values.'

The returns in a series may be correlated not with their immediately prior returns, but with more distant prior returns. Also, the portfolio returns—which are essentially averages of individual security returns—are strongly positively autocorrelated, even though the individual asset returns are negatively autocorrelated. This means large positive cross-correlations between assets across time (Campbell *et al.* 1997). Samuelson (1991) studies the effects of these intertemporal correlations on portfolio selection in a simple binomial model setup.

In this paper, the covariance matrices $\mathrm{Cov}_{j,n}$, with $j \neq n$, $j = 1, 2, \ldots, M+1$, $n = 1, 2, \ldots, M+1$, incorporate all the above-mentioned intertemporal correlations between the asset prices.

(b) *Assets follow geometric Brownian motion*

In this section, we derive solutions for the optimal trading strategy when assets follow geometric Brownian motion. The dynamics of each risky asset Z_i are given by (Dixit & Pindyck 1994)

$$\mathrm{d}Z_i = a_i Z_i(t)\,\mathrm{d}t + b_i Z_i(t)\,\mathrm{d}w_i,$$

where a_i is the constant conditional expected relative change in price per unit time, b_i^2 is the constant conditional variance per unit time, and $\mathrm{d}w_i$ is an increment of a Weiner process that is normally distributed with zero mean, that is, $E(\mathrm{d}w) = 0$. The processes $w_i(t)$ and $w_j(t)$ have correlated increments and we denote $\mathrm{Cov}\{b_i\,\mathrm{d}w_i(t), b_j\,\mathrm{d}w_j(t)\} = b_{i,j}\,\mathrm{d}t$. The expected prices and covariances of risky assets are generated by

$$\begin{aligned}
\mu_{i,j} &= E(Z_i)(t = t_j) = S_{i,0}\mathrm{e}^{a_i t_j},\\
\mathrm{Cov}_{j,n}(i,k) &= E\{[Z_i(t_j) - E[Z_i(t_j)]][Z_k(t_n) - E[Z_k(t_n)]]\}\\
&= S_{i,0}S_{k,0}\mathrm{e}^{a_i t_j + a_k t_n}\{\mathrm{e}^{b_{i,k}\min(t_j,t_n)} - 1\},
\end{aligned}$$

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where $S_{i,0}$ denotes the initial price of the ith asset and $b_{i,i} = b_i^2$. The risk-free asset is characterized by $S_{0,j} = S_{0,0}\mathrm{e}^{rt_j}$, where r is the risk-free interest rate. Substituting these into (3.5) and solving (3.10), we can get the optimal trading strategy. In what follows, we derive the optimal time-zero allocations to the risky assets, as $\Delta t \to 0$, by considering the two-period case. The first-order optimality conditions (3.10) can be written as ($S_{0,0} = 1$)

$$\frac{1}{S_{0,2}^2}\operatorname{Cov}_{2,2}(\alpha_0 + \alpha_1) - \frac{1}{S_{0,2}S_{0,1}}\operatorname{Cov}_{2,1}\alpha_1 = \theta\frac{X}{S_{0,0}}R_0 \tag{3.16}$$

and

$$\left\{\frac{1}{S_{0,2}^2}\operatorname{Cov}_{2,2} - \frac{1}{S_{0,2}S_{0,1}}\operatorname{Cov}_{1,2}\right\}(\alpha_0 + \alpha_1) + \left\{\frac{1}{S_{0,1}^2}\operatorname{Cov}_{1,1} - \frac{1}{S_{0,2}S_{0,1}}\operatorname{Cov}_{2,1}\right\}\alpha_1 = \theta\frac{X}{S_{0,0}}R_1. \tag{3.17}$$

Setting $t_1 = \Delta t$ and $t_2 = 2\Delta t$, and letting $\Delta t \to 0$, we get

$$\frac{1}{X}(\alpha_{1,0}S_{1,0}, \alpha_{2,0}S_{2,0}, \ldots, \alpha_{N,0}S_{N,0})^{\mathrm{T}} = \theta\Omega^{-1}(a_1 - r, a_2 - r, \ldots, a_N - r)^{\mathrm{T}} \tag{3.18}$$

and

$$\frac{1}{X}(\alpha_{1,1}S_{1,0}, \alpha_{2,1}S_{2,0}, \ldots, \alpha_{N,1}S_{N,0})^{\mathrm{T}} = 0,$$

where

$$\Omega = \begin{pmatrix} b_1^2 & b_{1,2} & \cdots & b_{1,N} \\ b_{2,1} & b_2^2 & \cdots & b_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ b_{N,1} & b_{N,2} & \cdots & b_N^2 \end{pmatrix}. \tag{3.19}$$

The time-zero allocations coincide with those obtained by Merton (1969) via the continuous-time framework, corresponding to a constant relative risk-averse investor.

(*c*) *Derivation of analytical solution when there is no intertemporal correlation between assets*

In this section, we derive the analytical solution for the trading strategy when there is no intertemporal correlation between any two assets at any two different trading dates, that is, when

$$\operatorname{Cov}_{j,n} = 0, \quad j \neq n, \quad j, n = 1, 2, \ldots, M+1.$$

Equations (3.7) and (3.8) simplify to

$$C_{0,0} = \frac{1}{S_{0,M+1}^2}\operatorname{Cov}_{M+1,M+1}, \quad C_{j,n} = \frac{1}{S_{0,M+1}^2}\operatorname{Cov}_{M+1,M+1}, \quad j \neq n,$$

and

$$C_{n,n} = \frac{1}{S_{0,M+1}^2}\operatorname{Cov}_{M+1,M+1} + \frac{1}{S_{0,n}^2}\operatorname{Cov}_{n,n}, \quad n \neq 0.$$

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Making these substitutions, the first-order optimality condition (3.10) can be written as ($S_{0,0} = 1$)

$$\frac{1}{S_{0,M+1}^2} \operatorname{Cov}_{M+1,M+1}\left\{\sum_{j=0}^{M} \alpha_j\right\} = \theta \frac{X}{S_{0,0}} R_0 \tag{3.20}$$

and

$$\frac{1}{S_{0,M+1}^2} \operatorname{Cov}_{M+1,M+1}\left\{\sum_{j=0}^{M} \alpha_j\right\} + \frac{1}{S_{0,k}^2} \operatorname{Cov}_{k,k}\, \alpha_k = \theta \frac{X}{S_{0,0}} R_k, \quad k = 1, 2, \ldots, M. \tag{3.21}$$

Solving the set of equations (3.20) and (3.21) and using $R_{M+1} = 0$, we get

$$\alpha_0 = \theta\left(\frac{X}{S_{0,0}}\right)\left\{\sum_{k=1}^{M+1} S_{0,k}^2 \operatorname{Cov}_{k,k}^{-1}(R_0 - R_k)\right\} \tag{3.22}$$

and

$$\alpha_k = -\theta\left(\frac{X}{S_{0,0}}\right) S_{0,k}^2 \operatorname{Cov}_{k,k}^{-1}(R_0 - R_k), \quad k = 1, 2, \ldots, M. \tag{3.23}$$

Let $U_j = (U_{1,j}, U_{2,j}, \ldots, U_{N,j})^{\mathrm{T}}$ denote the vector of the optimum number of shares invested in risky assets at time $t = t_j$, $j = 0, 1, \ldots, M$. Noting that

$$U_j = \sum_{k=0}^{j} \alpha_k, \quad j = 0, 1, \ldots, M,$$

we get

$$U_j = \theta\left(\frac{X}{S_{0,0}}\right)\left\{\sum_{k=j+1}^{M+1} S_{0,k}^2 \operatorname{Cov}_{k,k}^{-1}(R_0 - R_k)\right\}. \tag{3.24}$$

At time $t = t_j$, $j = 1, 2, \ldots, M$, if the assets are not correlated, we have

$$\operatorname{diag}(\operatorname{Cov}_{j,j}^{-1}) = \frac{1}{\sigma_i^2(j)},$$

where $\sigma_i^2(j)$, $i = 1, 2, \ldots, N$, denotes the variance of the ith asset at time $t = t_j$. In this case, equation (3.24) reduces to

$$U_{i,j} = \theta\left(\frac{X}{S_{0,0}}\right)\left[\sum_{k=j+1}^{M+1} \frac{E\{S_{i,k}/S_{0,k} - S_{i,0}/S_{0,0}\}}{\operatorname{Var}\{S_{i,k}/S_{0,k} - S_{i,0}/S_{0,0}\}}\right], \quad i = 1, 2, \ldots, N, \quad j = 0, 1, \ldots, M. \tag{3.25}$$

The optimal trading strategy corresponding to the risk-free asset can be obtained from the conservation-of-wealth equations.

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4. All controls are deterministic, short sales allowed

In the previous section, we maximized the net rate of profit subject to the budget constraint at time zero and subject to the conservation of the values of the portfolio at future trading dates during rebalancing. Since short sales are allowed in all the assets, the money borrowed from the bank has to be stochastic to preserve the conservation of wealth. In reality, short sales and random borrowing from the bank may not be possible. Also, if the portfolio does not contain a risk-free asset, it may not be possible to arrive at a neat rate of return R by solving the wealth conservation explicitly, as we did in the last section. If either of these two conditions is violated, wealth will not be conserved in future trading dates and the conservation of the wealth at future trading dates has to be weakened. In this section, we derive the Markowitz mean-variance principle for the allocation problem when the condition—'conservation of wealth'—at future trading dates is relaxed and when there are no short-sale restrictions on assets. This relaxation of wealth conservation will enable us to expand our relaxed formulation to derive the corresponding Markowitz mean-variance principle when there are short-sale restrictions, which is the subject of our study in the subsequent sections. Our objectives are as follows.

(1) To maximize the expected value of $W(t_{M+1})/\mu_{R,M+1}$ given by

$$\frac{W(t_{M+1})}{\mu_{R,M+1}} = \sum_{i=0}^{N} \frac{S_{i,M+1}}{\mu_{R,M+1}} U_{i,M}. \tag{4.1}$$

(2) To minimize the variance of $W(t_{M+1})/\mu_{R,M+1}$.

(3) To satisfy the budget constraint at time zero, that is, $\phi_0 = 0$, where

$$\phi_0 = \sum_{i=0}^{N} \frac{S_{i,0}}{\mu_{R,0}} U_{i,0} - \frac{X}{\mu_{R,0}}. \tag{4.2}$$

(4) To conserve the wealth at future trading dates in the expected sense, that is, $E(\phi_j) = 0$, where

$$\phi_j = \sum_{i=0}^{N} \frac{S_{i,j}}{\mu_{R,j}} (U_{i,j} - U_{i,j-1}), \quad j = 1, 2, \ldots, M. \tag{4.3}$$

(5) To minimize the variance of ϕ_j, $j = 1, 2, \ldots, M$.

In the above, $\mu_{R,j}$, $j = 0, 1, \ldots, M+1$, is the reference price of an asset at time $t = t_j$. Our objective (4) says that, at future trading dates, the value of the portfolio will be conserved in the expected sense during rebalancing. Objective (5) says that the deviation of ϕ_j from its mean is minimized, which is equivalent to saying that $E(\phi_j^2)$ is minimized.

This is a multi-objective equality-constrained optimization problem. Let

$$U_j = (U_{0,j}, U_{1,j}, \ldots, U_{N,j})^{\mathrm{T}}$$

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denote the $(N+1)$-dimensional vector of the number of shares held at time $t = t_j$. In the previous sections, this notation was used to denote the N-dimensional vector of the number of the shares. For brevity, we use the same notation. Let

$$\mu_j = \frac{1}{\mu_{R,j}}(\mu_{0,j}, \mu_{1,j}, \ldots, \mu_{N,j})^{\mathrm{T}}.$$

With this notation, we have

$$E\left\{\frac{W(t_{M+1})}{\mu_{R,M+1}}\right\} \equiv \mu_p = \mu_{M+1}^{\mathrm{T}} U_M, \tag{4.4}$$

$$\mathrm{Var}\left\{\frac{W(t_{M+1})}{\mu_{R,M+1}}\right\} \equiv \sigma_p^2 = \frac{1}{\mu_{R,M+1}^2} U_M^{\mathrm{T}} \,\mathrm{Cov}_{M+1,M+1}\, U_M. \tag{4.5}$$

With the implicit notation of $\mu_{i,0} = S_{i,0}$, $i = 0, 1, \ldots, N$, the budget constraint $\phi_0 = 0$ leads to

$$\mu_0^{\mathrm{T}} U_0 = \frac{X}{\mu_{R,0}} \equiv \bar{X} \tag{4.6}$$

and the constraint $E(\phi_j) = 0$ leads to

$$\mu_j^{\mathrm{T}}(U_j - U_{j-1}) = 0, \quad j = 1, 2, \ldots, M. \tag{4.7}$$

Similarly, we can derive

$$\mathrm{Var}(\phi_j) \equiv \sigma_{\phi_j}^2 = \frac{1}{\mu_{R,j}^2}(U_j - U_{j-1})^{\mathrm{T}} \,\mathrm{Cov}_{j,j}(U_j - U_{j-1}). \tag{4.8}$$

Our Markowitz mean-variance problem is to find the dynamic trading strategy $\{U_j,\ j = 0, 1, \ldots, M\}$ that minimizes

$$F = -\theta' \mu_{M+1}^{\mathrm{T}} U_M + \frac{1}{2}\left\{\frac{1}{\mu_{R,M+1}^2} U_M^{\mathrm{T}} \,\mathrm{Cov}_{M+1,M+1}\, U_M + \sum_{j=1}^{M}\left[\frac{\theta_j}{\mu_{R,j}^2}(U_j - U_{j-1})^{\mathrm{T}} \,\mathrm{Cov}_{j,j}(U_j - U_{j-1})\right]\right\}, \tag{4.9}$$

subject to the budget constraint (4.6) and the expected wealth conservation equations given by (4.7). In (4.9), θ' is the parameter that defines the risk aversion of the investor and θ_j (greater than 0), $j = 1, 2, \ldots, M$, are *a priori* given parameters that quantify how the wealth is conserved at future trading dates. Define

$$\frac{\theta_j}{\mu_{R,j}^2} = \psi_j, \quad j = 1, 2, \ldots, M, \quad \frac{1}{\mu_{R,M+1}^2} = \psi_{M+1}, \tag{4.10}$$

$U = (U_0^{\mathrm{T}}, U_1^{\mathrm{T}}, \ldots, U_M^{\mathrm{T}})^{\mathrm{T}}$ and the matrix C as shown in Appendix A.

It can be verified that

$$\sum_{j=1}^{M} \psi_j (U_j - U_{j-1})^{\mathrm{T}} \,\mathrm{Cov}_{j,j}(U_j - U_{j-1}) + \psi_{M+1} U_M^T \,\mathrm{Cov}_{M+1,M+1}\, U_M = U^{\mathrm{T}} C U. \tag{4.11}$$

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Defining

$$A = \begin{pmatrix} \mu_0^{\mathrm{T}} & 0 & 0 & & \cdots & 0 \\ -\mu_1^{\mathrm{T}} & \mu_1^{\mathrm{T}} & 0 & & \cdots & 0 \\ 0 & -\mu_2^{\mathrm{T}} & \mu_2^{\mathrm{T}} & & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & & -\mu_M^{\mathrm{T}} & \mu_M^{\mathrm{T}} \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} \bar{X} \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

so that $b \in R^{M+1}$, equations (4.6) and (4.7) can together be cast in the form

$$AU = b. \tag{4.12}$$

Defining $P = (0^{\mathrm{T}}, 0^{\mathrm{T}}, \ldots, \mu_{M+1}^{\mathrm{T}})^{\mathrm{T}} \in R^{(M+1)(N+1)}$, we can write the expected value of the portfolio as

$$\mu_{M+1}^{\mathrm{T}} U_M = P^{\mathrm{T}} U. \tag{4.13}$$

With this notation, our Markowitz problem is to minimize

$$F = -\theta' P^{\mathrm{T}} U + \tfrac{1}{2} U^{\mathrm{T}} C U, \tag{4.14}$$

subject to the equality constraint (4.12). From (4.11) and the way in which the matrix C is composed, we know that C is always symmetric and positive-semi-definite and positive-definite if all covariance matrices $\mathrm{Cov}_{i,i}$ are positive-definite. The constraint matrix A contains the expected prices of the assets at different times, and it is always of full-row rank.

We form the Lagrangian

$$L = -\theta' P^{\mathrm{T}} U + \tfrac{1}{2} U^{\mathrm{T}} C U + \lambda^{\mathrm{T}} \{AU - b\}, \tag{4.15}$$

where $\lambda = (\lambda_0, \lambda_1, \ldots, \lambda_M)^{\mathrm{T}} \in R^{M+1}$ is a vector of multipliers associated with the equality constraints (4.12). The first-order optimality condition gives

$$CU + A^{\mathrm{T}} \lambda = \theta' P, \tag{4.16}$$

$$AU = b. \tag{4.17}$$

Equations (4.16) and (4.17) resemble exactly those of Markowitz (1987) and in what follows we derive analytical solutions when C is positive-definite on the one hand and positive-semi-definite on the other.

(*a*) *C is positive-definite*

We assume that all covariance matrices $\mathrm{Cov}_{j,j}$ are positive-definite, which means that all assets are risky and no asset can be written as a linear combination of the rest. Solving (4.16), we get

$$U = \theta' C^{-1} P - C^{-1} A^{\mathrm{T}} \lambda. \tag{4.18}$$

Substituting (4.18) into (4.17), we arrive at

$$G\lambda = \theta' A C^{-1} P - b, \tag{4.19}$$

where $G = AC^{-1}A^{\mathrm{T}}$ is a positive-definite matrix, since C is positive-definite and A is of full-row rank. Solving (4.19), we obtain

$$\lambda = R^1 + \theta' R^2, \qquad U = H^1 + \theta' H^2, \tag{4.20}$$

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where

$$R^1 = -G^{-1}b, \qquad R^2 = G^{-1}AC^{-1}P, \tag{4.21}$$

$$H^1 = C^{-1}A^{\mathrm{T}}G^{-1}b, \qquad H^2 = C^{-1}P - C^{-1}A^{\mathrm{T}}G^{-1}AC^{-1}P, \tag{4.22}$$

$$H^2 = (H_0^{2^{\mathrm{T}}}, H_1^{2^{\mathrm{T}}}, \ldots, H_M^{2^{\mathrm{T}}})^{\mathrm{T}}, \qquad H^1 = (H_0^{1^{\mathrm{T}}}, H_1^{1^{\mathrm{T}}}, \ldots, H_M^{1^{\mathrm{T}}})^{\mathrm{T}},$$

$$R^2 = (R_0^2, R_1^2, \ldots, R_M^2)^{\mathrm{T}}, \qquad R^1 = (R_0^1, R_1^1, \ldots, R_M^1)^{\mathrm{T}}.$$

In the above, H_j^1, H_j^2, $j = 0, 1, \ldots, M$, are $(N+1)$-dimensional vectors and R^1, R^2 are $(M+1)$-dimensional vectors. Equation (4.20) states that when other parameters θ_j, $j = 1, 2, \ldots, M$, are fixed, the optimal portfolio is linear in the risk-aversion parameter θ'. Substituting the expressions for U and λ into (4.16) and (4.17), we have

$$CH^1 + A^{\mathrm{T}}R^1 = 0, \qquad AH^1 = b, \tag{4.23}$$

$$CH^2 + A^{\mathrm{T}}R^2 = P, \qquad AH^2 = 0, \tag{4.24}$$

from which we obtain

$$H^{2^{\mathrm{T}}}CH^1 = 0, \qquad H^{2^{\mathrm{T}}}CH^2 = H^{2^{\mathrm{T}}}P = \mu_{M+1}^{\mathrm{T}}H_M^2. \tag{4.25}$$

(*b*) *Efficient frontier*

Substituting the optimal portfolio given by (4.20) into (4.13), we get the optimal return of the portfolio given by

$$\mu_p = k_0 + \theta' k_1, \tag{4.26}$$

where

$$k_0 = \mu_{M+1}^{\mathrm{T}}H_M^1, \qquad k_1 = \mu_{M+1}^{\mathrm{T}}H_M^2. \tag{4.27}$$

Now we obtain the total risk, defined by the variance of the terminal wealth and the square of the deviation from the expected return during trading dates, that is,

$$\text{Risk} = \sigma_{\mathrm{tot}}^2 = U^{\mathrm{T}}CU. \tag{4.28}$$

It can be shown that

$$\sigma_{\mathrm{tot}}^2 = \beta_0 + \theta'^2\beta_2, \tag{4.29}$$

where $\beta_0 = H^{1^{\mathrm{T}}}CH^1$, $\beta_2 = H^{2^{\mathrm{T}}}CH^2 = k_1$. Eliminating θ' from (4.26) and (4.29), we obtain the efficient frontier

$$\frac{\sigma_{\mathrm{tot}}^2 - \beta_0}{\beta_2} = \left(\frac{\mu_p - k_0}{k_1}\right)^2. \tag{4.30}$$

The efficient frontier given by (4.30) is a parabola in $(\sigma_{\mathrm{tot}}^2, \mu_p)$-space and is a hyperbola in $(\sigma_{\mathrm{tot}}, \mu_p)$-space. It can be seen that the total risk explicitly contains the parameters θ_j, $j = 1, 2, \ldots, M$. In practice, we need only to know the total risk with $\theta_j = 1$, $j = 1, 2, \ldots, M$, and this is considered in § 5 *b*, where the Markowitz mean-variance principle is illustrated by a numerical example when short sales are not allowed.

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(*c*) *C is positive-semi-definite*

In this section, we derive the optimum trading strategy when there exists a risk-free asset, identified by the index 0. We take the reference price $\mu_{R,j}$ to be $S_{0,j}$, $j = 0, 1, \ldots, M+1$. In this case, the covariance matrix $\mathrm{Cov}_{j,j}$ is given by

$$\mathrm{Cov}_{j,j} = \begin{pmatrix} 0 & 0 \\ 0 & D_{j,j} \end{pmatrix}, \tag{4.31}$$

where $D_{j,j}$ is the $N \times N$ covariance matrix associated with the risky assets at a trading date $t = t_j$. We introduce notation so that the problem can be cast in a compact from (see Appendix B). The first-order optimality conditions (4.16) and (4.17) lead to

$$A_1^{\mathrm{T}} \lambda = \theta' h_1, \tag{4.32}$$

$$DV + A_2^{\mathrm{T}} \lambda = \theta' h_2, \tag{4.33}$$

$$A_1 T_0 + A_2 V = b. \tag{4.34}$$

Solving (4.32), we obtain

$$\lambda = \theta' l, \tag{4.35}$$

where $l = (1, 1, \ldots, 1)^{\mathrm{T}}$ is an $(M+1)$-dimensional vector containing ones. Since no risky asset can be written as a linear combination of the rest of risky assets, the $D_{j,j}$ are positive-definite and hence D is positive-definite. Substituting (4.35) into (4.33) and (4.34), we get

$$V = \theta' \{D^{-1} h_2 - D^{-1} A_2^{\mathrm{T}} l\}, \tag{4.36}$$

$$T_0 = A_1^{-1} b - \theta' A_1^{-1} A_2 D^{-1} \{h_2 - A_2^{\mathrm{T}} l\}. \tag{4.37}$$

For $\theta' = 0$, we get

$$V = 0, \qquad T_0 = A_1^{-1} b = \frac{X}{\mu_{0,0}} l. \tag{4.38}$$

Simplifying, equation (4.36) can be written as

$$V = \theta' D^{-1} (g_1^{\mathrm{T}} - g_0^{\mathrm{T}}, g_2^{\mathrm{T}} - g_1^{\mathrm{T}}, \ldots, g_{M+1}^{\mathrm{T}} - g_M^{\mathrm{T}})^{\mathrm{T}}, \tag{4.39}$$

which, using the inverse of the matrix given by (A 1), can be written as

$$V = \theta' \left(\left[\sum_{j=1}^{M+1} \frac{D_{j,j}^{-1}}{\psi_j} (g_j - g_0) \right]^{\mathrm{T}}, \ldots, \left[\sum_{j=M+1}^{M+1} \frac{D_{j,j}^{-1}}{\psi_j} (g_j - g_0) \right]^{\mathrm{T}} \right)^{\mathrm{T}}. \tag{4.40}$$

In terms of the expected prices, the above equation can be written as

$$V_k = \theta' \sum_{j=k+1}^{M+1} \frac{\mu_{0,j}^2}{\theta_j} D_{j,j}^{-1} \left(\frac{\mu_{1,j}}{\mu_{0,j}} - \frac{\mu_{1,0}}{\mu_{0,0}}, \frac{\mu_{2,j}}{\mu_{0,j}} - \frac{\mu_{2,0}}{\mu_{0,0}}, \ldots, \frac{\mu_{N,j}}{\mu_{0,j}} - \frac{\mu_{N,0}}{\mu_{0,0}} \right)^{\mathrm{T}}, \quad k = 0, 1, \ldots, M. \tag{4.41}$$

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Substituting the expressions for $A_1^{-1}A_2$ (see Appendix B), equation (4.37) leads to

$$T_0 = \begin{pmatrix} U_{0,0} \\ U_{0,1} \\ U_{0,2} \\ \vdots \\ U_{0,M} \end{pmatrix}$$

$$= \frac{X}{\mu_{0,0}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} - \theta' \begin{pmatrix} \displaystyle\sum_{j=1}^{M+1} g_0^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) \\ \displaystyle\sum_{j=1}^{M+1} g_0^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) - \sum_{j=1}^{1} g_j^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) \\ \displaystyle\sum_{j=1}^{M+1} g_0^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) - \sum_{j=1}^{2} g_j^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) \\ \vdots \\ \displaystyle\sum_{j=1}^{M+1} g_0^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) - \sum_{j=1}^{M} g_j^{\mathrm{T}} \frac{D_{j,j}^{-1}}{\psi_j}(g_j - g_0) \end{pmatrix}. \tag{4.42}$$

If the assets are not correlated at trading dates $t = t_j$, $j = 1, 2, \ldots, M$, we have

$$\operatorname{diag}(D_{j,j}^{-1}) = \frac{1}{\sigma_i^2(j)},$$

where $\sigma_i^2(j)$, $i = 1, 2, \ldots, N$, denotes the variance of the ith stock at time $t = t_j$. In this case, equation (4.41) further reduces to

$$U_{i,j} = \theta' \sum_{k=j+1}^{M+1} \frac{1}{\theta_k} \frac{E\{S_{i,k}/S_{0,k} - S_{i,0}/S_{0,0}\}}{\operatorname{Var}\{S_{i,k}/S_{0,k} - S_{i,0}/S_{0,0}\}}. \tag{4.43}$$

It may be noted that for $\theta'/\theta_k = \theta(X/S_{0,0})$ equations (3.25) and (4.43) coincide.

5. No short sales allowed

In this section, we study the allocation problem when short sales are not allowed. The Markowitz problem is to minimize

$$F = -\theta' P^{\mathrm{T}} U + \tfrac{1}{2} U^{\mathrm{T}} C U \tag{5.1}$$

subject to the equality constraints given by (4.12) and the non-negativity constraints on the trading strategy U. The first-order KKT optimality conditions are

$$CU + A^{\mathrm{T}}\lambda = \theta' P + Y, \tag{5.2}$$

$$AU = b, \tag{5.3}$$

$$-U \leqslant 0 \tag{5.4}$$

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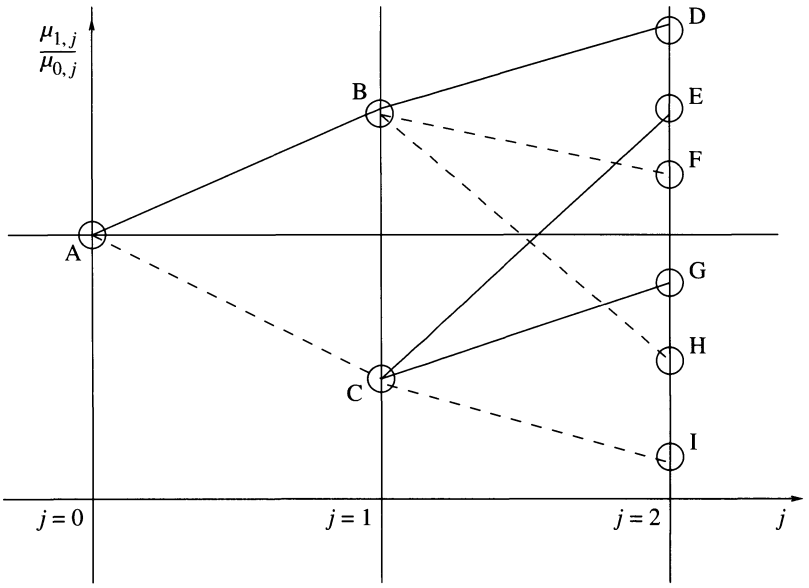


Figure 1. Scenarios for the expected price of the risky asset with respect to the risk-free asset for the two-period two-asset case test problem.

and

$$U_{i,j}Y_{i,j} = 0, \quad i = 0, 1, \ldots, N, \quad j = 0, 1, \ldots, M, \tag{5.5}$$

$$Y_{i,j} \geqslant 0, \quad i = 0, 1, \ldots N, \quad j = 0, 1, \ldots, M, \tag{5.6}$$

where

$$Y = (Y_0^{\mathrm{T}}, Y_1^{\mathrm{T}}, \ldots, Y_M^{\mathrm{T}})^{\mathrm{T}}, \quad Y_j = (Y_{0,j}, Y_{1,j}, \ldots, Y_{N,j})^{\mathrm{T}}, \quad j = 0, 1, \ldots, M.$$

Here, Y_j is a vector of multipliers associated with the non-negativity constraints on vector U_j, and the vector Y contains the collection of all vectors of multipliers.

The structure of the problem remains exactly the same as that of the Markowitz single-period allocation problem (Markowitz 1987). There are no closed-form solutions for general problems. The existing computer code for quadratic programming problems can be used to obtain numerical solutions. The algorithms may be enriched by using the special structure of the matrix C as done by Markowitz & Perold (1989); however, this is not dealt with here. Following Best & Grauer (1991*a*, *b*), the sensitivity of the optimal portfolio due to changes in mean prices of the assets may be studied. In the next section, we derive closed-form solutions when there is only one risky asset and a risk-free asset, and when the number of trading dates is two. In the subsequent section, we illustrate the Markowitz mean-variance principle by solving a test problem numerically using the revised simplex method.

(*a*) *Analytical solution for a two-period two-assets case*

In this section we obtain analytical solutions for the trading strategy when the portfolio consists of one risky asset and a risk-free asset. The number of periods under consideration is two. We consider all possible price movements of the risky

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asset, as shown in figure 1. Figure 1 shows the ratio between the expected price of the risky asset and the price of the unit risk-free asset as a function of the trading time. In figure 1, point A corresponds to $\mu_{1,0}/\mu_{0,0}$, points B and C correspond to $\mu_{1,1}/\mu_{0,1}$ and points D, E, F, G, H, and I correspond to $\mu_{1,2}/\mu_{0,2}$.

Between two trading dates, the price history of the risky asset is unknown, though it is shown by a linear line in figure 1. Let $\sigma_1^2(1)$ and $\sigma_1^2(2)$ denote the variances of the risky asset at trading dates t_1 and t_2, respectively. For this simple problem, we have

$$C = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \psi_1\sigma_1^2(1) & 0 & -\psi_1\sigma_1^2(1) \\ 0 & 0 & 0 & 0 \\ 0 & -\psi_1\sigma_1^2(1) & 0 & \psi_1\sigma_1^2(1) + \psi_2\sigma_1^2(2) \end{pmatrix},$$

$$b = \left(\frac{X}{\mu_{0,0}}, 0\right)^{\mathrm{T}},$$

$$A = \begin{pmatrix} 1 & \dfrac{\mu_{1,0}}{\mu_{0,0}} & 0 & 0 \\ -1 & -\dfrac{\mu_{1,1}}{\mu_{0,1}} & 1 & \dfrac{\mu_{1,1}}{\mu_{0,1}} \end{pmatrix},$$

$$P = \left(0, 0, 1, \frac{\mu_{1,2}}{\mu_{0,2}}\right)^{\mathrm{T}},$$

$$U = (U_{0,0}, U_{1,0}, U_{0,1}, U_{1,1})^{\mathrm{T}},$$

$$Y = (Y_{0,0}, Y_{1,0}, Y_{0,1}, Y_{1,1})^{\mathrm{T}},$$

$$\lambda = (\lambda_0, \lambda_1)^{\mathrm{T}}.$$

First we obtain analytical solutions for the KKT conditions (5.2)–(5.6) when no inequality constraint is active. This implies that we can set

$$Y_{0,0} = Y_{1,0} = Y_{0,1} = Y_{1,1} = 0. \tag{5.7}$$

Due to this selection, equations (5.5) and (5.6) are automatically satisfied. Solving (5.2) and (5.3), we get

$$\lambda_0 = \lambda_1 = \theta', \tag{5.8}$$

$$U_{0,0} = \frac{X}{\mu_{0,0}} - \theta'\frac{\mu_{1,0}}{\mu_{0,0}}\left[\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\} + \frac{1}{\psi_2\sigma_1^2(2)}\left\{\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right], \tag{5.9}$$

$$U_{1,0} = \theta'\left[\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\} + \frac{1}{\psi_2\sigma_1^2(2)}\left\{\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right], \tag{5.10}$$

$$U_{0,1} = U_{0,0} + \theta'\left[\frac{\mu_{1,1}}{\mu_{0,1}}\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right],$$

$$U_{1,1} = \theta'\left[\frac{1}{\psi_2\sigma_1^2(2)}\left\{\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right]. \tag{5.11}$$

The satisfaction of inequality constraints, given by (5.4), depends upon the expected price of the risky asset relative to the price of the risk-free asset. In what follows, we consider all possible price movements of the risky asset.

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(*b*) *Case 1*

We consider the case when

$$\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \geqslant 0, \qquad \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \geqslant 0. \tag{5.12}$$

Conditions (5.12) correspond to the price situations ABD and ABF in figure 1. When conditions (5.12) are satisfied, $U_{1,1} \geqslant 0$, $U_{1,0} \geqslant 0$, $U_{0,1} \geqslant U_{0,0}$ and $U_{0,0} \geqslant 0$, if $\theta' \leqslant L_1$, where

$$L_1 = \frac{X}{\mu_{0,0}} \Bigg/ \left(\frac{\mu_{1,0}}{\mu_{0,0}} \left[\frac{1}{\psi_1 \sigma_1^2(1)} \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} + \frac{1}{\psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} \right] \right). \tag{5.13}$$

For $0 \leqslant \theta' \leqslant L_1$, all KKT conditions (5.2)–(5.6) are satisfied, $U_{0,0}(\theta' = L_1) = 0$, and all the money is allocated to the risky asset when $\theta' = L_1$.

(*c*) *Case 2*

We consider the case when

$$\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \geqslant 0, \qquad \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \leqslant 0 \tag{5.14}$$

and

$$\frac{1}{\psi_1 \sigma_1^2(1)} \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} + \frac{1}{\psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} \geqslant 0. \tag{5.15}$$

Conditions (5.14) and (5.15) correspond to the price situation ACE in figure 1. When conditions (5.14) and (5.15) are satisfied, $U_{1,1} \geqslant 0$, $U_{1,0} \geqslant 0$, $U_{0,0} \geqslant U_{0,1}$ and $U_{0,1} \geqslant 0$, if $\theta' \leqslant L_2$. Here,

$$L_2 = \frac{X/\mu_{0,0}}{L_2^*}, \tag{5.16}$$

where

$$L_2^* = \frac{\mu_{1,0}}{\mu_{0,0}} \left[\frac{1}{\psi_1 \sigma_1^2(1)} \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} + \frac{1}{\psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} \right] - \frac{\mu_{1,1}}{\mu_{0,1}} \frac{1}{\psi_1 \sigma_1^2(1)} \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\}.$$

For $0 \leqslant \theta' \leqslant L_2$, all KKT conditions (5.2)–(5.6) are satisfied, $U_{0,1}(\theta' = L_2) = 0$, and $U_{0,0}(\theta' = L_2)$ is positive. It can be noted that even if $\mu_{1,1}/\mu_{0,1} - \mu_{1,0}/\mu_{0,0} \leqslant 0$, there are situations when we have investment in the risky asset. Similarly, we can find the optimal trading strategy for $\theta' > L_2$, but to save space it is not dealt with here.

(*d*) *Case 3*

We consider the case when

$$\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \leqslant 0, \qquad \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \geqslant 0. \tag{5.17}$$

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Conditions (5.17) correspond to the price situation ABH in figure 1. Equation (5.11) says that $U_{1,1}$ will be negative when the first condition in (5.17) is satisfied. Let

$$U_{1,1} = 0, \qquad Y_{0,0} = Y_{0,1} = Y_{1,0} = 0, \tag{5.18}$$

so that (5.5) is satisfied. Solving the KKT conditions (5.2) and (5.3) along with (5.18), we get

$$\begin{aligned}
\lambda_0 &= \lambda_1 = \theta', \\
U_{0,0} &= \frac{X}{\mu_{0,0}} - \theta'\left[\frac{\mu_{1,0}}{\mu_{0,0}}\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right], \\
U_{1,0} &= \theta'\left[\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right], \\
U_{0,1} &= \frac{X}{\mu_{0,0}} + \theta'\left[\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}^2\right], \\
Y_{1,1} &= -\theta'\left\{\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}.
\end{aligned}$$

When conditions (5.17) are satisfied, $U_{1,0} \geqslant 0$, $U_{0,1} \geqslant 0$, $Y_{1,1} \geqslant 0$ and $U_{0,0} \geqslant 0$, if $\theta' \leqslant L_3$, where

$$L_3 = \frac{X}{\mu_{0,0}} \Big/ \left(\frac{\mu_{1,0}}{\mu_{0,0}}\left[\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\}\right]\right).$$

For $0 \leqslant \theta' \leqslant L_3$, all KKT conditions (5.2)–(5.6) are satisfied, and $U_{0,0}(\theta' = L_3) = 0$, and all the money is allocated to the risky asset when $\theta' = L_3$.

(e) *Case 4*

We consider the case when

$$\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,1}}{\mu_{0,1}} \geqslant 0, \qquad \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \leqslant 0 \tag{5.19}$$

and

$$\frac{1}{\psi_1\sigma_1^2(1)}\left\{\frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\} + \frac{1}{\psi_2\sigma_1^2(2)}\left\{\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}}\right\} \leqslant 0. \tag{5.20}$$

Conditions (5.19) and (5.20) correspond to the price situation ACG in figure 1. Equation (5.10) says that $U_{1,0}$ will be negative when condition (5.20) is satisfied. Let

$$U_{1,0} = 0, \qquad Y_{0,0} = Y_{0,1} = Y_{1,1} = 0, \tag{5.21}$$

so that (5.5) is satisfied. Solving the KKT conditions (5.2) and (5.3), along with (5.21), we get

$$\lambda_0 = \lambda_1 = \theta',$$

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$$U_{0,0} = \frac{X}{\mu_{0,0}},$$

$$U_{0,1} = \frac{X}{\mu_{0,0}} - \theta' \left[\frac{\mu_{1,1}}{\mu_{0,1}} \frac{1}{\psi_1 \sigma_1^2(1) + \psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,1}}{\mu_{0,1}} \right\} \right],$$

$$U_{1,1} = \theta' \left[\frac{1}{\psi_1 \sigma_1^2(1) + \psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,1}}{\mu_{0,1}} \right\} \right],$$

$$Y_{1,0} = -\theta' \frac{\psi_1 \sigma_1^2(1) \psi_2 \sigma_1^2(2)}{\psi_1 \sigma_1^2(1) + \psi_2 \sigma_1^2(2)} \times \left[\frac{1}{\psi_1 \sigma_1^2(1)} \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} + \frac{1}{\psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\} \right].$$

When conditions (5.19) and (5.20) are satisfied, $U_{1,1} \geqslant 0$, $Y_{1,0} \geqslant 0$ and $U_{0,1} \geqslant 0$, if $\theta' \leqslant L_4$, where

$$L_4 = \frac{X}{\mu_{0,0}} \Big/ \left(\frac{\mu_{1,1}}{\mu_{0,1}} \left[\frac{1}{\psi_1 \sigma_1^2(1) + \psi_2 \sigma_1^2(2)} \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,1}}{\mu_{0,1}} \right\} \right] \right).$$

For $0 \leqslant \theta' \leqslant L_4$, all KKT conditions (5.2)–(5.6) are satisfied. It may be noted that, for all $0 \leqslant \theta' \leqslant L_4$, even though the final return and variance are different, the allocation $U_{0,0}$ remains the same. Hence it is optimal to select $\theta' = 0$, and the initial allocation is independent of preferences.

(*f*) *Case 5*

We consider the final obvious case when

$$\frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,0}}{\mu_{0,0}} \leqslant 0, \qquad \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \leqslant 0. \tag{5.22}$$

Conditions (5.22) correspond to the price situation ACI in figure 1. Equations (5.10) and (5.11) say that $U_{1,0}$ and $U_{1,1}$ will be negative when condition (5.22) is satisfied. Let

$$U_{1,0} = U_{1,1} = 0, \qquad Y_{0,0} = Y_{1,0} = 0, \tag{5.23}$$

so that (5.5) is satisfied. Solving the KKT conditions (5.2) and (5.3), along with (5.23), we get

$$\lambda_0 = \lambda_1 = \theta',$$

$$U_{0,0} = U_{0,1} = \frac{X}{\mu_{0,0}},$$

$$Y_{1,0} = -\theta' \left\{ \frac{\mu_{1,1}}{\mu_{0,1}} - \frac{\mu_{1,0}}{\mu_{0,0}} \right\},$$

$$Y_{1,1} = -\theta' \left\{ \frac{\mu_{1,2}}{\mu_{0,2}} - \frac{\mu_{1,1}}{\mu_{0,1}} \right\}.$$

When (5.22) is satisfied, all KKT conditions are satisfied and all the money is deposited only in the risk-free asset at all times.

For fixed parameters θ_j, $j = 1, 2, \ldots, M$, it is seen that the optimal trading strategy is piecewise linear in the risk parameter θ' when short sales are not allowed. The bounds for the risk parameters L_1, L_2, L_3 and L_4 depend linearly on the initial wealth X, implying that the wealthier the investor the more risk the investor takes.

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(g) Numerical example—seven assets and ten periods

In this section, we illustrate the generalized Markowitz mean-variance principle by solving numerically a hypothetical asset-allocation problem when the portfolio consists of six risky assets and a risk-free asset, and when the number of trading periods is ten. We assume that the dynamics of the risky asset Z_i, $i = 2, 3, \ldots, 7$, follows a generalized uncorrelated geometric Brownian motion given by

$$\frac{\mathrm{d}Z_i}{Z_i} = a_i(t)\,\mathrm{d}t + b_i(t)\,\mathrm{d}w, \tag{5.24}$$

where $a_i(t)$ is the conditional expected relative change in price per unit time, b_i^2 is the conditional variance per unit time and $\mathrm{d}w$ is the increment of a Weiner process. The risk-free asset Z_1 is governed by

$$\frac{\mathrm{d}Z_1}{Z_1} = r\,\mathrm{d}t, \tag{5.25}$$

where r denotes the risk-free interest rate.

The probability density function $f(Z_i, t)$, which characterizes the dynamics of the asset Z_i, $i = 2, 3, \ldots, 7$, is obtained by solving the Fokker–Planck equation (Dixit & Pindyck 1994)

$$\frac{\partial f}{\partial t}(Z_i, t) + \frac{\partial}{\partial Z_i}[a_i(t) Z_i f(Z_i, t)] - \frac{1}{2}\frac{\partial^2}{\partial Z_i^2}[b_i^2(t) Z_i^2 f(Z_i, t)] = 0, \tag{5.26}$$

along with the initial condition

$$f(Z_i, t = 0) = \delta(Z_i - S_{i,0}), \tag{5.27}$$

where $S_{i,0}$ denotes the initial unit price of the ith asset and δ denotes the Dirac delta function. The expected price and the variance of each stock Z_i, by integrating the Fokker–Planck equation (5.26), are given by

$$\left.\begin{aligned} E\left[\frac{Z_i}{S_{i,0}}\right] &= \exp\left(\int_0^t a_i(y)\,\mathrm{d}y\right), \\ \mathrm{Var}\left[\frac{Z_i}{S_{i,0}}\right] &= \exp\left(\int_0^t 2a_i(y)\,\mathrm{d}y\right)\left\{\exp\left(\int_0^t b_i^2(y)\,\mathrm{d}y\right) - 1\right\}. \end{aligned}\right\} \tag{5.28}$$

For our numerical illustration, we assume that the drift and diffusion coefficients are of the form

$$a_i(t) = a_{i,0} + a_{i,1}t, \quad b_i(t) = b_i, \quad i = 2, 3, \ldots, 7.$$

The expected prices and variances of risky assets are generated by

$$\mu_{i,j} = E[Z_i](t = t_j), \quad \sigma_i^2(j) = \mathrm{Var}[Z_i](t = t_j), \quad i = 2, 3 \ldots, 7.$$

From (5.28), it can be seen that

$$\xi_i(t) \equiv \frac{\mathrm{d}}{\mathrm{d}t}\left\{E\left[\frac{Z_i}{S_{i,0}}\right]\right\} = \mathrm{e}^{a_{i,0}t + a_{i,1}t^2/2}(a_{i,0} + a_{i,1}t),$$

and we have the following four different types of behaviour for the expected price of the risky asset.

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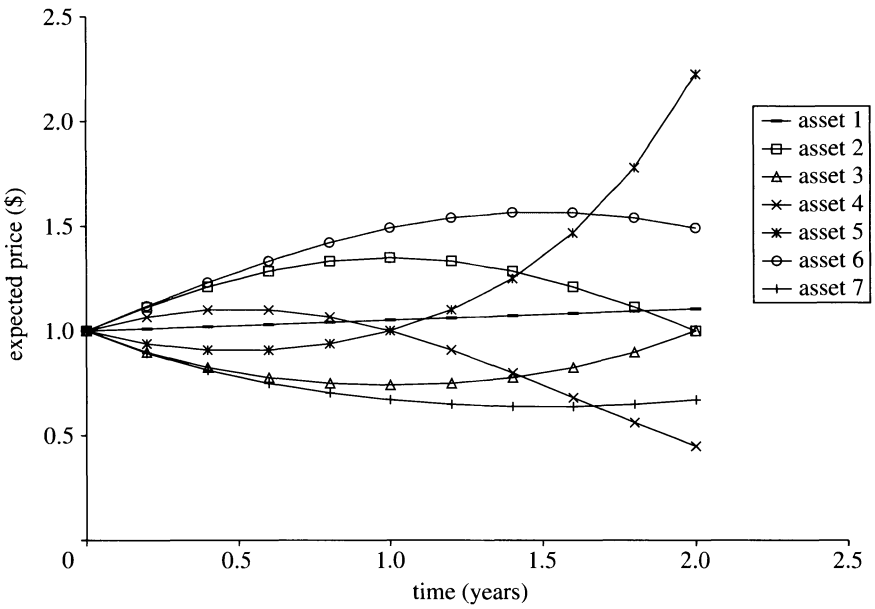


Figure 2. Expected prices of the assets for the test problem that has six risky assets and one risk-free asset.

(1) For $a_{i,0} > 0$, $a_{i,1} > 0$, $\xi_i(t) > 0$ for all t.

(2) For $a_{i,0} > 0$, $a_{i,1} < 0$, $\xi_i(t) > 0$ for $0 < t < -a_{i,0}/a_{i,1}$, and $\xi_i(t) < 0$ for $t > -a_{i,0}/a_{i,1}$.

(3) For $a_{i,0} < 0$, $a_{i,1} > 0$, $\xi_i(t) < 0$ for $0 < t < -a_{i,0}/a_{i,1}$, and $\xi_i(t) > 0$ for $t > -a_{i,0}/a_{i,1}$.

(4) For $a_{i,0} < 0$, $a_{i,1} < 0$, $\xi_i(t) < 0$ for all t.

To simulate all these possible cases, six risky assets are considered and the corresponding numerical values for the parameters $a_{i,0}$, $a_{i,1}$ and b_i are as shown in table 1, the risk-free interest rate being $r = 0.05$. For uniformity, we have taken the initial price of each asset $S_{i,0}$ to be unity. We have imposed a limit of 11 trading dates ($M = 9$) and limited the duration of investment to $T = 2$ yr. The expected prices of all the assets and their variances are shown in figures 2 and 3. We assume the initial wealth X to be \$100 and set the weight parameters ψ_i, $i = 1, 2, \ldots, M + 1$, to unity.

We present the numerical results obtained by solving the quadratic programming problem using the revised simplex method. The quadratic programming algorithm needs to be run until the entire initial wealth is invested in a single asset that has the largest return at time zero. Figures 4–10 show the surface plots for the number of shares held in respective assets as a function of time and the total risk, where the total risk is defined as

$$\sigma_{\mathrm{T}}^2 = \text{total risk} = U^{\mathrm{T}} C^* U,$$

where matrix C^* is obtained from matrix C by setting $\psi_j = 1$, $j = 1, 2, \ldots, M + 1$. This total risk is a measure and is explicitly independent of the weight parameters ψ_j, though the trading strategy vector U depends upon them. Figure 11 shows the

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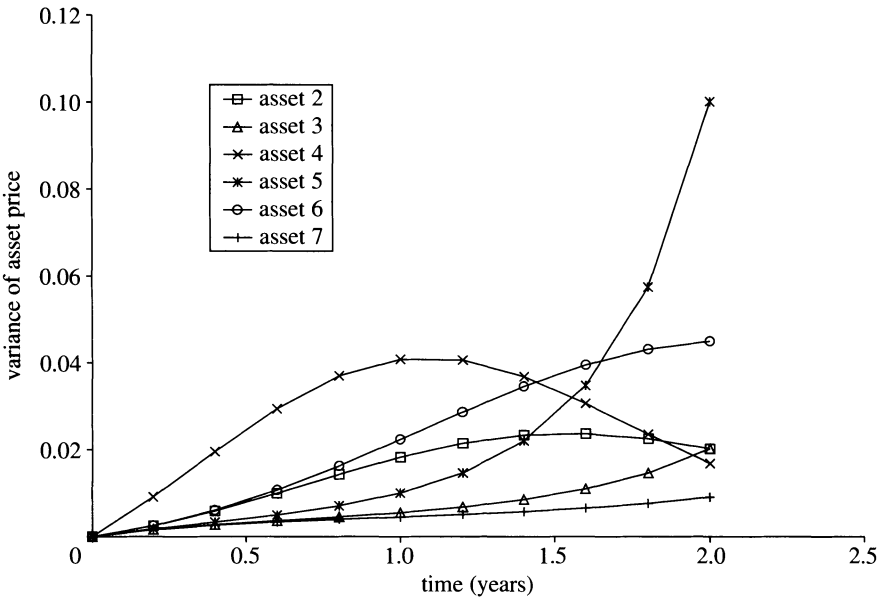


Figure 3. Variances of the prices of the risky assets for the test problem that has six risky assets and one risk-free asset.

Table 1. *Numerical values for the parameters* $a_{i,0}$, $a_{i,1}$, b_i, $i = 2, 3, \ldots, 7$, *for the test problem that has six risky assets and one risk-free asset*

risky asset number, i	$a_{i,0}$	$a_{i,1}$	b_i
2	0.6	−0.6	0.1
3	−0.6	0.6	0.1
4	0.4	−0.8	0.2
5	−0.4	0.8	0.1
6	0.6	−0.4	0.1
7	−0.6	0.4	0.1

time-zero allocations to the assets as a function of total risk, and figure 12 shows the optimal allocations at the end of the period as a function of total risk.

Figure 2 shows the expected prices of the assets as a function of time. Initially, assets 1, 4, 2 and 6 have increasing expected asset prices, and assets 5, 3, and 7 have decreasing expected asset prices in that order. Beyond $t = 0.5$, $t = 1.0$ and $t = 1.5$ yr, assets 5, 3 and 7, respectively, have increasing expected asset prices. Similarly, beyond $t = 0.5$, $t = 1.0$ and $t = 1.5$ yr, assets 4, 2 and 6 have decreasing expected asset prices, respectively. At the terminal period, the expected price of asset 2 dips below the unit price of the risk-free asset, whereas the expected price of asset 6 is still above the unit price of the risk-free asset. It can be expected that there are allocations to assets 1, 4, 2 and 6 and no allocation to assets 5, 3 and 7 at time $t = 0$ and allocations to assets 5, 3 and 7 beyond $t = 0.5$, $t = 1.0$ and $t = 1.5$ yr, respectively.

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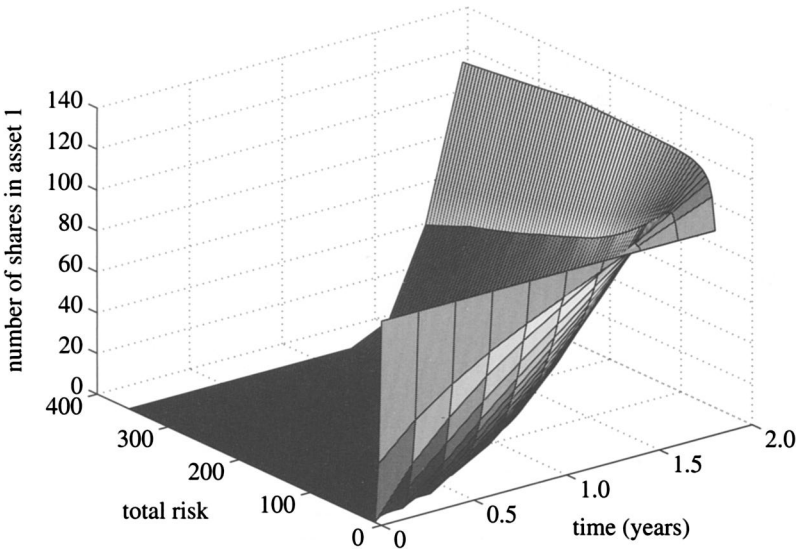


Figure 4. Number of shares in asset 1 (risk free) with an initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

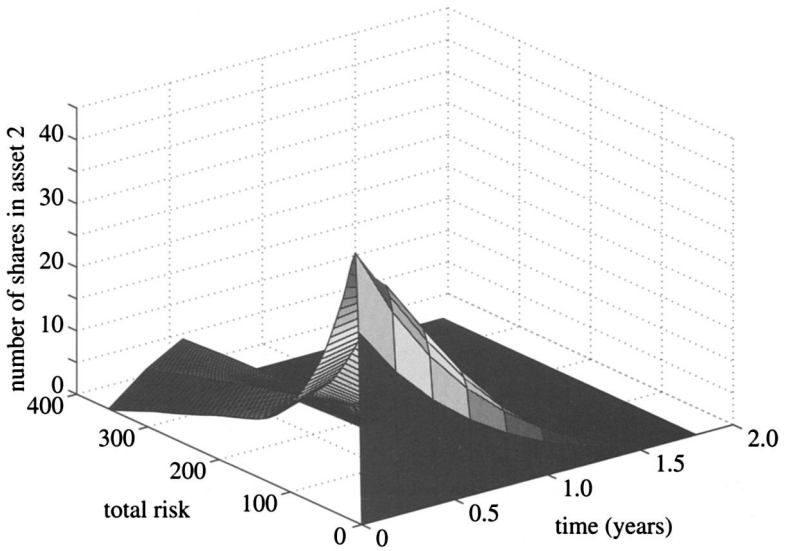


Figure 5. Number of shares in asset 2 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

Figures 4 and 11 show that, for zero total risk, everything is allocated to the risk-free asset at all times, and figure 11 shows that there is no allocation to assets 3, 5 and 7 at time zero and that as the total risk increases from zero the allocations to assets 4, 2 and 6 increase. Beyond certain total risk (σ_{T}^2), there is no allocation to the risk-free asset and followed by the asset 4 at time zero. As total risk increases further, the allocation in asset 2 gradually decreases, and eventually at $\sigma_{\mathrm{T}}^2 = 355.0$, everything is allocated to asset 6.

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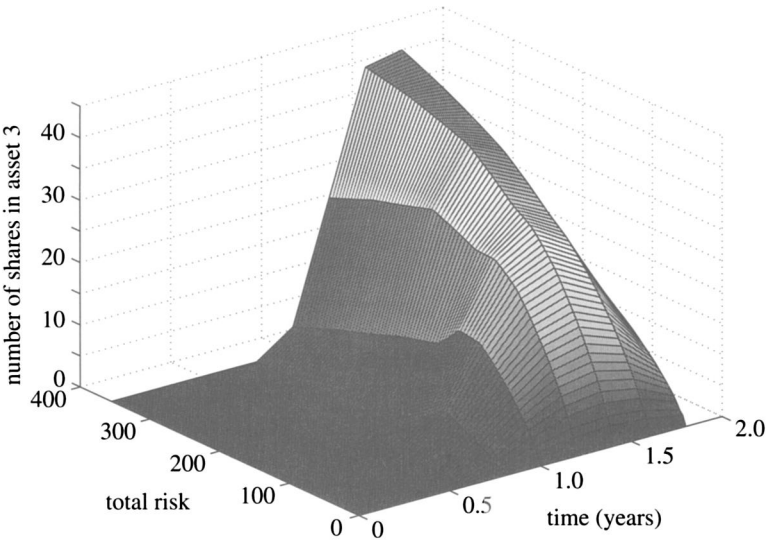


Figure 6. Number of shares in asset 3 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

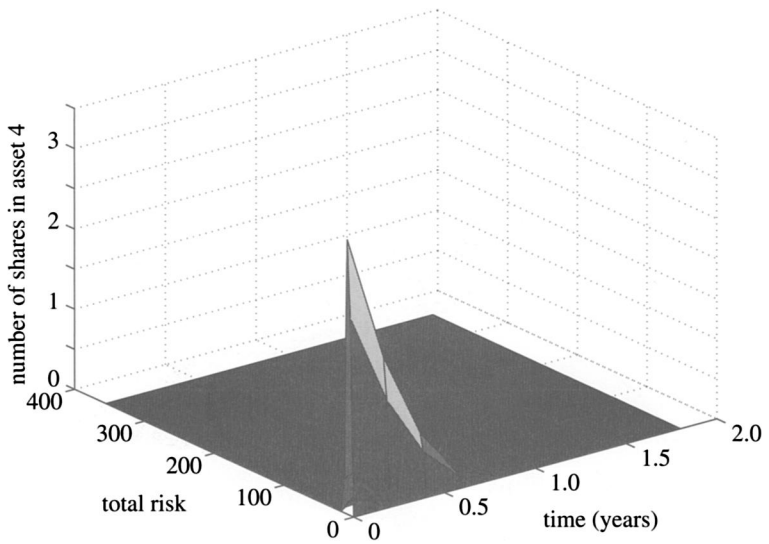


Figure 7. Number of shares in asset 4 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

Figures 5, 7 and 12 show that there is no allocation to assets 2 and 4 at the terminal time $t = T$. It can be seen (see figure 12) that even at the highest total risk, a maximum amount is allocated to the risk-free asset followed by risky assets 5, 3, 7 and 6. The surface plot for the number of shares allocated to the risk-free asset, depicted by figure 4, shows that there is increasing allocation to the risk-free asset at future trading dates, even when there is no allocation to risk-free asset at time zero. The surface plot for the number of shares allocated to asset 2, depicted by figure 5,

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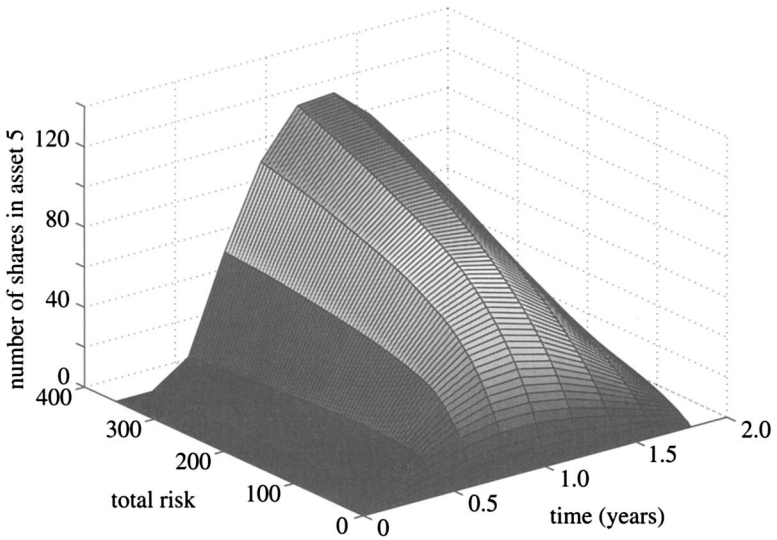


Figure 8. Number of shares in asset 5 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

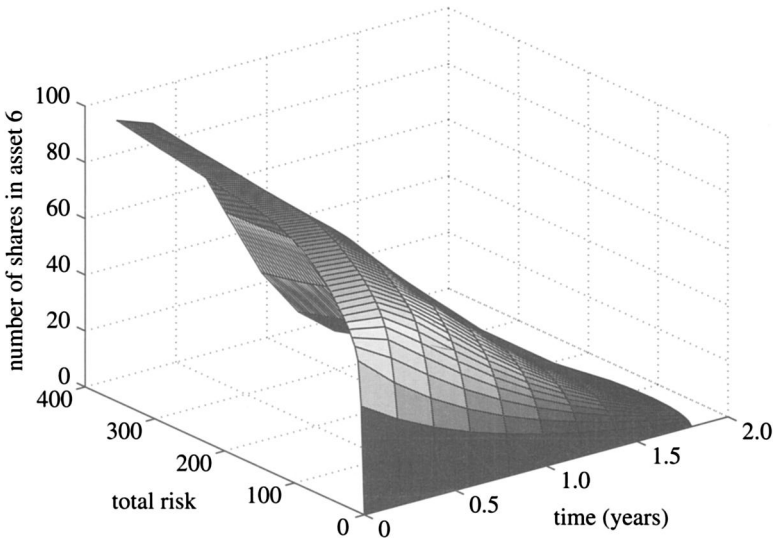


Figure 9. Number of shares in asset 6 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

shows that the allocation increases as a function of total risk and then decreases during initial periods of investment and that there is no allocation after $t = 1.5$ yr.

The expected prices of assets 3 and 5 demonstrate similar behaviour, and hence the assets have similar allocation patterns, as shown in figures 6 and 8, though they are different in magnitude. Since the expected price of asset 4 increases marginally over a small period of time (see figure 2), there is marginal allocation to it initially and no allocation to it at later times as the expected price dips far below the unit price of the risk-free asset, as shown in figure 7. Asset 6 has the largest expected

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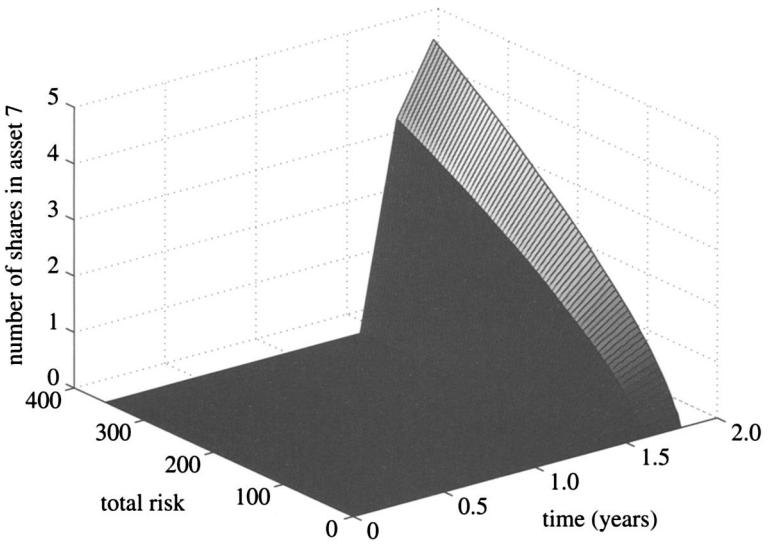


Figure 10. Number of shares in asset 7 with the initial wealth of $X = \$100$ and initial asset prices $S_{i,0} = \$1.0$, $i = 1, 2, \ldots, 7$.

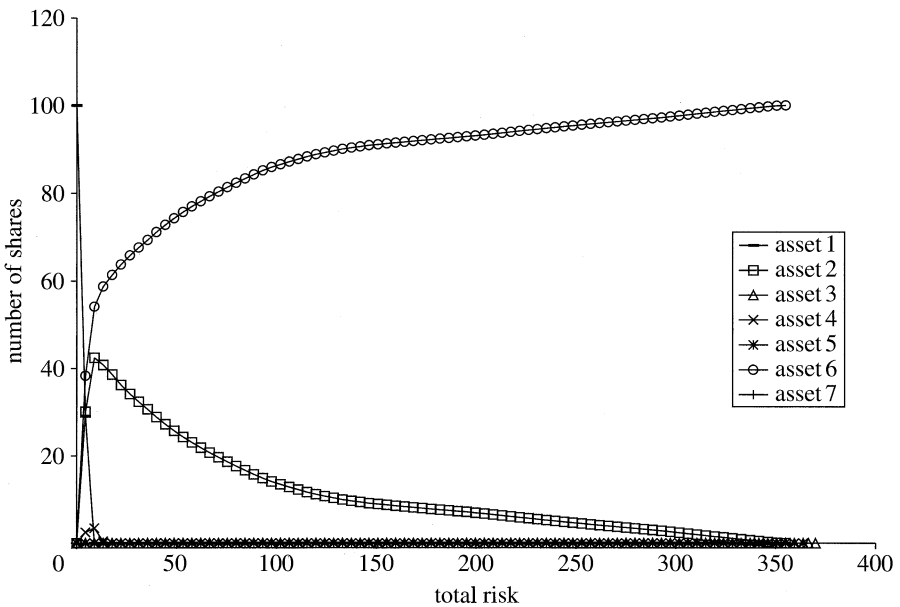


Figure 11. Time-zero allocations to the assets for the test problem that has six risky assets and a risk-free asset.

price initially and hence has the maximum time-zero allocation corresponding to the maximum total risk. The allocation to asset 6, as shown in figure 9, decreases gradually as a function of time. The expected price of asset 7 decreases, except near the terminal period (see figure 2), and hence asset 7 has non-zero allocation, as shown in figure 10, after time $t = 1.5$ yr. Figures 13 and 14 show the efficient frontier in $(\mu_R, \sigma_{\mathrm{T}}^2)$- and $(\mu_R, \sigma_{\mathrm{T}})$-space, respectively.

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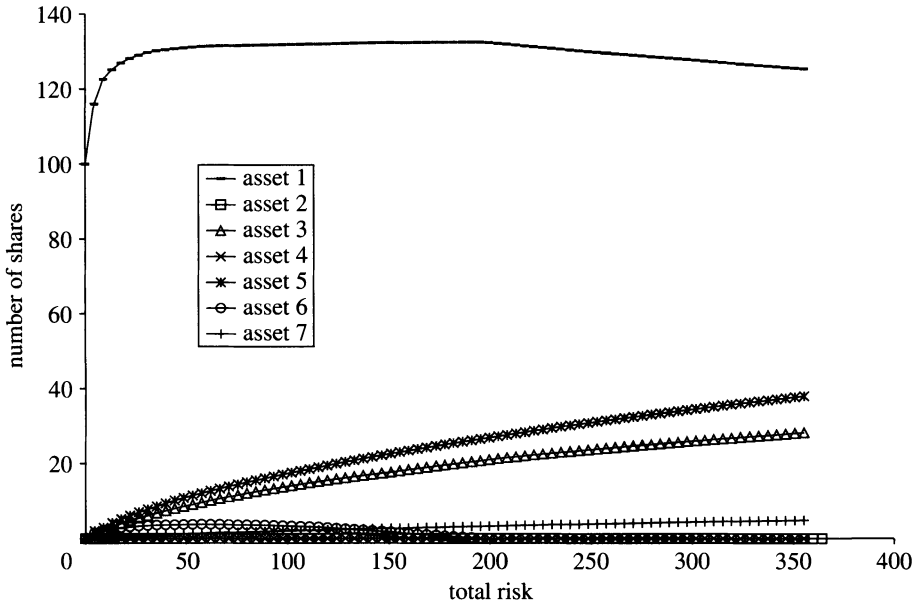


Figure 12. Terminal time allocations to the assets for the test problem that has six risky assets and a risk-free asset.

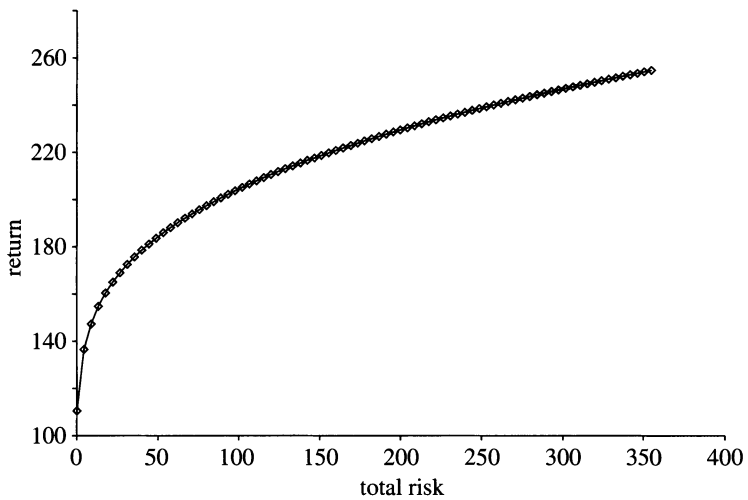


Figure 13. Efficient frontier in the return/total variance space for the test problem that has six risky assets and one risk-free asset.

6. Implications of the assumptions on the results

In this section, we discuss the implications of the assumptions, especially assumption (a) in § 2 *a*, on the results.

We have assumed that the number of shares invested in each risky asset is deterministic at all future trading dates. In principle, the number of shares to be transacted at future trading dates should be a non-anticipative process of the underlying stochastic processes of the asset prices. When the trading strategy is dynamic and

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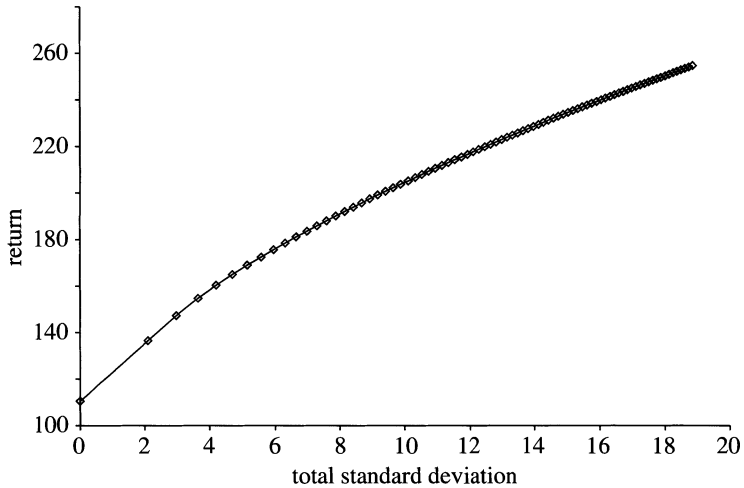


Figure 14. Efficient frontier in the return/total standard deviation space for the test problem that has six risky assets and one risk-free asset.

dependent on the asset prices, it is possible to 'buy low and sell high' by using volatility of the stocks in a pumping action (Luenberger 1998; Fernholz & Shay 1982). This can be achieved by expressing the trading strategy as a function of the asset prices in an appropriate fashion so that the model avoids looking into the future. While this feedback policy (Kendrick 1981) enhances the performance of the portfolio, it requires more information about the statistical properties of the asset prices.

On the contrary, the open-loop dynamic policy, as stipulated in condition (a) in § 2 *a*, requires up to second-order moments of the asset prices. The efficient frontier at the end of the planning horizon provides recommendations that are different from the single-period model. The top of the efficient frontier has a mixture of many assets (see figure 12), rather than the typical single asset, coming out of the single-period optimizer. This difference has been observed by Mulvey (2000) as well. Moreover, due to the dynamic nature of the trading strategy, the model incorporates the intertemporal correlations between assets, so that the 'predictability of the asset prices' is used optimally. This is not the case in the single-period model.

In the absence of transaction and market impact costs, dynamic and state-dependent strategies can improve the performance of the portfolio by taking advantage of the volatility of the risky assets in order to gain greater return. However, this requires that the trading strategy be asset-price dependent, so that the policy is feedback. In this paper, we have considered only dynamic open-loop policy and the volatility pumping is not considered. Transaction and market impact costs will dampen the advantage due to volatility pumping, and further research is needed to find the optimal rebalancing strategy.

7. Conclusion

The dynamic portfolio-selection problem is formulated as a Markowitz mean-variance optimization problem. It is shown that the single-period Markowitz quadratic programming algorithm can be used to solve the multi-period asset-allocation problem with suitable modifications in the covariance and linear constraint matrices. It is

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assumed that the number of shares invested in risky assets is deterministic and that the amount of money invested in the risk-free asset is random at future trading dates when short sales on assets are allowed. The general covariance matrix in the multi-period setup contains intertemporal correlations between assets, in addition to correlations between assets at all trading dates. Analytical solutions are obtained for the optimal trading strategy, which is linear in the risk-aversion parameter, implying that any efficient portfolio can be generated by two efficient portfolios. The efficient frontier is a straight line in the expected return/standard deviation of the portfolio space. When the dynamics of the risky assets follow geometric Brownian motion, it is shown that the time-zero allocations to the risky assets coincide with those obtained by Merton in the continuous-time framework. When short sales are not allowed on assets, the values of the portfolio at future trading dates may not be conserved. In the modified Markowitz mean-variance formulation, the value of the portfolio at future trading dates is conserved in the expected sense, and the optimal trading strategy is selected so that the deviation is minimized in the least-square sense. It is shown that the efficient frontier is a parabola in the expected return/total variance space when short sales are allowed and when the portfolio consists of risky assets only. When short sales are not allowed, it is shown that finding the optimal trading strategy is equivalent to solving the single-period Markowitz quadratic programming problem with the modifications in the covariance and linear constraint matrices. By solving the KKT conditions, analytical solutions are obtained for a two-period two-assets case and the Markowitz mean-variance principle is illustrated by solving a test problem numerically using the revised simplex method. The top of the efficient frontier has a mixture of many assets rather than the typical single asset coming out of the single-period optimizer. The special structure of the generalized covariance matrix C may be exploited to enhance the speed of the resulting quadratic programming algorithm, though it has not been done here.

The authors thank the referees and Professor Michael J. Best, University of Waterloo, for constructive suggestions. The first author is greatly indebted to his teacher Professor George D. Raithby, Department of Mechanical Engineering, University of Waterloo.

Appendix A.

This appendix defines the covariance matrix C used in § 4. The covariance matrix C is given by

$$C = \begin{pmatrix} C_{0,0} & C_{0,1} & 0 & 0 & \cdots & 0 \\ C_{1,0} & C_{1,1} & C_{1,2} & 0 & \cdots & 0 \\ 0 & C_{2,1} & C_{2,2} & C_{2,3} & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & C_{M,M-1} & C_{M,M} \end{pmatrix}.$$

Here, each submatrix $C_{i,j}$ of C is an $(N+1) \times (N+1)$ matrix defined by

$$C_{0,0} = \psi_1 \operatorname{Cov}_{1,1}, \quad C_{0,1} = -\psi_1 \operatorname{Cov}_{1,1}, \quad C_{0,k} = 0, \quad k = 2, 3, \ldots, M,$$

and for all $i = 1, 2, \ldots, M-1$,

$$C_{i,i} = \psi_i \operatorname{Cov}_{i,i} + \psi_{i+1} \operatorname{Cov}_{i+1,i+1}, \quad C_{i,i-1} = -\psi_i \operatorname{Cov}_{i,i},$$
$$C_{i,i+1} = -\psi_{i+1} \operatorname{Cov}_{i+1,i+1}, \quad C_{i,k} = 0, \quad k \leqslant i-2, \quad k \geqslant i+2,$$

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and

$$C_{M,M} = \psi_M \operatorname{Cov}_{M,M} + \psi_{M+1} \operatorname{Cov}_{M+1,M+1},$$
$$C_{M,M-1} = -\psi_M \operatorname{Cov}_{M,M}, \quad C_{M,k} = 0, \quad k \leqslant M-2.$$

It is not necessary to invert C when all covariance matrices $\operatorname{Cov}_{j,j}$ are positive-definite. Its inverse is given by

$$C^{-1} = \begin{pmatrix} \sum_{j=1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \sum_{j=2}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \cdots & \sum_{j=M+1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} \\ \sum_{j=2}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \sum_{j=2}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \cdots & \sum_{j=M+1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=M+1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \sum_{j=M+1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} & \cdots & \sum_{j=M+1}^{M+1} \dfrac{\operatorname{Cov}_{j,j}^{-1}}{\psi_j} \end{pmatrix}. \tag{A 1}$$

Appendix B.

This appendix details the notation used in § 4 *c*. Let

$$U_j = (U_{0,j}, V_j^{\mathrm{T}})^{\mathrm{T}}, \quad V_j = (U_{1,j}, U_{2,j}, \ldots, U_{N,j})^{\mathrm{T}}, \qquad j = 0, 1, \ldots, M,$$
$$\mu_j = (1, g_j^{\mathrm{T}})^{\mathrm{T}}, \qquad g_j = \left(\frac{\mu_{1,j}}{\mu_{0,j}}, \frac{\mu_{2,j}}{\mu_{0,j}}, \ldots, \frac{\mu_{N,j}}{\mu_{0,j}} \right)^{\mathrm{T}}, \quad j = 0, 1, \ldots, M+1,$$
$$T_0 = (U_{0,0}, U_{0,1}, \ldots, U_{0,M})^{\mathrm{T}}, \qquad V = (V_0^{\mathrm{T}}, V_1^{\mathrm{T}}, \ldots, V_M^{\mathrm{T}})^{\mathrm{T}},$$
$$h_1 = (0, 0, \ldots, 1)^{\mathrm{T}} \in R^{M+1}, \qquad h_2 = (0^{\mathrm{T}}, 0^{\mathrm{T}}, \ldots, g_{M+1}^{\mathrm{T}})^{\mathrm{T}} \in R^{(M+1)N}$$

and

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ -1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & -1 & 1 \end{pmatrix},$$

$$D = \begin{pmatrix} C_{0,0} & C_{0,1} & 0 & 0 & \cdots & 0 \\ C_{1,0} & C_{1,1} & C_{1,2} & 0 & \cdots & 0 \\ 0 & C_{2,1} & C_{2,2} & C_{2,3} & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & C_{M,M-1} & C_{M,M} \end{pmatrix},$$

$$A_2 = \begin{pmatrix} g_0^{\mathrm{T}} & 0 & 0 & 0 & \cdots & 0 \\ -g_1^{\mathrm{T}} & g_1^{\mathrm{T}} & 0 & 0 & \cdots & 0 \\ 0 & -g_2^{\mathrm{T}} & g_2^{\mathrm{T}} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & -g_M^{\mathrm{T}} & g_M^{\mathrm{T}} \end{pmatrix},$$

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where each submatrix $C_{i,j}$ of D is an $N \times N$ matrix defined by

$$C_{0,0} = \psi_1 D_{1,1}, \quad C_{0,1} = -\psi_1 D_{1,1}, \quad C_{0,k} = 0, \quad k = 2, 3, \ldots, M,$$

and for all $i = 1, 2, \ldots, M-1$,

$$C_{i,i} = \psi_i D_{i,i} + \psi_{i+1} D_{i+1,i+1}, \quad C_{i,i-1} = -\psi_i D_{i,i},$$
$$C_{i,i+1} = -\psi_{i+1} D_{i+1,i+1}, \quad C_{i,k} = 0, \quad k \leqslant i-2, \quad k \geqslant i+2,$$

and

$$C_{M,M} = \psi_M D_{M,M} + \psi_{M+1} D_{M+1,M+1}, \quad C_{M,M-1} = -\psi_M D_{M,M},$$
$$C_{M,k} = 0, \quad k \leqslant M-2.$$

In the above, A_1 is an $(M+1) \times (M+1)$ matrix, T_0 is an $(M+1)$-dimensional vector that defines the allocation to the risk-free asset at all $M+1$ trading dates and V is a vector that contains the unknown N-dimensional vectors of the number of shares allocated to the risky assets. The inverse of the matrix A_1 and $A_1^{-1}A_2$ are, respectively, given by

$$A_1^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 & 0 \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{pmatrix},$$

$$A_1^{-1}A_2 = \begin{pmatrix} g_0^{\mathrm{T}} & 0 & 0 & 0 & \cdots & 0 \\ g_0^{\mathrm{T}} - g_1^{\mathrm{T}} & g_1^{\mathrm{T}} & 0 & 0 & \cdots & 0 \\ g_0^{\mathrm{T}} - g_1^{\mathrm{T}} & g_1^{\mathrm{T}} - g_2^{\mathrm{T}} & g_2^{\mathrm{T}} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & g_{M-1}^{\mathrm{T}} & 0 \\ g_0^{\mathrm{T}} - g_1^{\mathrm{T}} & g_1^{\mathrm{T}} - g_2^{\mathrm{T}} & g_2^{\mathrm{T}} - g_3^{\mathrm{T}} & \cdots & g_{M-1}^{\mathrm{T}} - g_M^{\mathrm{T}} & g_M^{\mathrm{T}} \end{pmatrix}.$$

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